**University Cheikh Anta Diop of Dakar**

**Faculty of Sciences and Techniques**

Department of Mathematics and Computer Science

Option: Algebra

Homological Algebra Master's Thesis

# Study of resolutions and homological dimensions in comp (A-Mod)

Presented by: Mouhamadou Moustapha Diop

Under the supervision of Professor Mohamed Ben Maaouia

Jury Members

| Name | First Name | Institution | Role |
|---|---|---|---|
| Sangharé | Mamadou | UCAD | President |
| Diankha | Oumar | UCAD | Examiner |
| Sow | Djiby | UCAD | Examiner |
| Barry | Mamadou | UCAD | Examiner |
| Maaouia | Mohamed Ben | UGB | Supervisor |

## Acknowledgements

My most sincere thanks to Professor Maaouia, for his assistance, availability, and devotion during the homological algebra course that served as the basis for this thesis. I also thank the LACGAA as a structure for providing the material and human resources that support this report. I also thank my classmates for the sustained motivation through continuous exchanges on our respective subjects.

To Professor Mohamed Ben Maaouia.

# Contents


**Introduction** **5**

**1 Modules, Basic Definitions** **7**
1.1 Tools . . . 7
1.1.1 Inductive and projective limit . . . 7
1.2 Operations on modules . . . 11
1.2.1 Direct sum . . . 11
1.2.2 Tensor product . . . 12
1.2.3 Short exact sequence of modules . . . 12
1.3 Some types of modules . . . 13
1.3.1 Free module . . . 14
1.3.2 Projective module . . . 14
1.3.3 Injective modules . . . 15
1.3.4 Flat module . . . 15
1.4 Bimodule . . . 16

**2 The category of complexes of $A$-modules : Comp $(A$-Mod$)$** **17**
2.1 Category and functors . . . 17
2.1.1 Category . . . 17
2.1.2 Functor . . . 18
2.1.3 Natural Transformations . . . 18
2.2 Category of complexes of A-module . . . 19
2.2.1 Complexes . . . 19
2.2.2 Homotopic chain complexes of A-module . . . 22
2.2.3 Homology of a complex of A-module . . . 23
2.2.4 The cone . . . 26

**3 Resolutions and homological dimensions in Comp $(A$-Mod$)$** **28**
3.1 Resolutions . . . 28
3.1.1 The functors Hom and $\otimes$ . . . 28
3.1.1.1 The functor Hom . . . 28
3.1.1.2 The functor $\otimes$ . . . 29
3.1.2 Projective and injective resolutions . . . 30
3.1.2.1 K-projective complex . . . 30

3.1.2.2 K-projective resolution . . . . . 32
3.1.3 Resolutions of unbounded complexes . . . . . 34
3.1.4 Flat resolution . . . . . 38
3.2 Homological dimensions . . . . . 38
3.2.1 The functors Ext and Tor . . . . . 38
3.2.1.1 The functor Ext . . . . . 38
3.2.1.2 The functor Tor . . . . . 39
3.2.2 The K-projective, K-injective and K-flat dimensions . . . 39

# Introduction

In this report, we explain in detail, the definitions and constructions needed to establish a proper homology theory in the category of complexes of modules of an associative ring. For this purpose, we follow the standard methods found in any homological algebra textbook. In our case, we rely principally on the course proposed by the Professor Maaouia, and on An introduction to homological algebra by Charles A Weibel [5]. For the construction of resolutions in the unbounded case, we follow the methods proposed by N. Spaltenstein in Resolutions of unbounded complexes[4]. Finally, in the last chapter we present some propositions on homological dimensions, using methods found in Homological dimensions of unbounded complexes[1]. The document is divided into chapters as follows :

In Chapter I, we recall some elementary notions of module theory. We omit most proofs but provide a reference where one can be found. We therefore define, among others things, projective, injective and flat modules, which are central notions to the subject, and somme basic tools needed for the construction of resolutions of unbounded complexes.

In Chapter II, we introduce the category of complexes of $A$-modules starting with category theory, which provides some concepts needed in that chapter and the following. We then go further, defining notions such as short exact sequences, cone, and related theorems and propositions.

In the third chapter, we define the category of complexes of $A$-module, and their resolutions and their dimensions, concluding with some properties of their homological dimensions.

In this document $A$ is an associative ring. Each time the term A-module is used, it will be a left $A$-module unless specified.

# Chapter 1

# Modules, Basic Definitions

In this chapter, we introduce direct and projective systems and their limits. We also discuss notions from modules theory that are essential for the development of the subject, and some types of modules (projective, injective, and flat modules).

## 1.1 Tools

### 1.1.1 Inductive and projective limit

**Definition 1.1.** A totally ordered set $I$ is said to be well-ordered when $\forall E \subseteq I$ such that $E \neq \emptyset$, $E$ has a least element ($\exists m \in E \mid \forall e \in E, m \leq e$).

**Proposition 1.1.** *Let $I$ be a well-ordered set, then:*

1. *$I$ has a least element,*
2. *if $i \in I$ is not the greatest element of $I$, then the set of elements greater than $i$ has a least element called the successor of $i$. And $i$ is the predecessor of its successor. And $i$ is the predecessor of its successor.*
3. *Every element of $I$ has at most one predecessor.*

**Definition 1.2.**

- Let $(I, \leq)$ be an ordered set, a direct system of $A$-modules is a sequence $(M_i)_{i \in I}$ of $A$-modules, and for all $i, j \in I$ such that $i \leq j$, a morphism $\phi^i_j : M_i \mapsto M_j$ such that:
  - $\forall i \in I$, $f^i_i = \mathrm{id}_{M_i}$,
  - $\forall i, j, k \in I$ such that $i \leq j \leq k$, $\phi^j_k \circ \phi^i_j = \phi^i_k$.
- We say that it is a projective system when for all $i, j \in I$ such that $i \leq j$, we have a morphism $\phi^j_i : M_j \mapsto M_i$ such that:
  - $\forall i \in I$, $f^i_i = \mathrm{id}_{M_i}$,

* $\forall i, j, k \in I$ such that $i \leq j \leq k$, $\phi_i^j \circ \phi_j^k = \phi_i^k$.

**Definition 1.3.**

- Given a direct system of $A$-modules $(M_i)_I$, the inductive limit of this system is another $A$-module $D$, and morphisms $\iota_i : M_i \mapsto D$ for each $i \in I$ such that for $i \leq j$, $\iota_j \circ \phi_j^i = \iota_i$.

$$\begin{tikzcd} M_i \arrow[rr, "\phi_j^i"] \arrow[dr, "\iota_i"'] & & M_j \arrow[dl, "\iota_j"] \\ & D & \end{tikzcd}$$

  Such that for any $A$-module $X$, and morphisms $f_i : M_i \mapsto X$ that satisfy $f_j \circ \phi_j^i = f_i$, there exists a unique morphism $u : D \mapsto X$ such that $\forall i \in I$, $f_i = u \circ \iota_i$.

- Similarly for a projective system of $A$-modules $(M_i)_I$, the projective limit is an $A$-module $L$ with morphisms $\pi_i : L \mapsto M_i$ for each $i \in I$ such that for all $\forall i \leq j$, $\phi_i^j \circ \pi_j = \pi_i$.

$$\begin{tikzcd} & L \arrow[dl, "\pi_j"'] \arrow[dr, "\pi_i"] & \\ M_j \arrow[rr, "\phi_i^j"] & & M_i \end{tikzcd}$$

  And for any $A$-module $X$, with morphisms $f_i : X \mapsto M_i$ that satisfy $\phi_i^j \circ f_j = f_i$, there exists a unique morphism $u : X \mapsto L$ such that $\forall i \in I$, $f_i = \pi_i \circ u$.

**Proposition 1.2.** *Let $(M_i)_I$ be a projective system of $A$-modules, then the $A$-module*

$$L = \left\{ (m_i) \in \prod_{i \in I} M_i \mid \forall i \leq j, m_i = \phi_i^j(m_j) \right\}$$

*is a limit of this projective system.*

*Proof.* Let $(m_i)_I, (m_i')_I \in L$; so $\forall i \leq j \in I$, $m_i = \phi_i^j(m_j)$ and $m_i' = \phi_i^j(m_j') \implies m_i + m_i' = \phi_i^j(m_j + m_j') \implies (m_i + m_i') \in L$. And for any $a \in A$, then $am_i = a\phi_i^j(m_j) = \phi_i^j(am_j)$ so $(am_i)_I \in L$. Thus $L$ is an $A$-module.

$$\begin{aligned} \pi_i : L &\longmapsto M_i \\ (m_i)_I &\longmapsto m_i \end{aligned}$$

$\forall (m_i)_I \in L$; $\phi_i^j \circ \pi_j\left((m_i)_I\right) = \phi_i^j(m_j) = m_i$. And let $X$, and the morphisms $f_i : X \mapsto M_i$ such that $\forall i \leq j \in I$, $\forall x \in X$; $\phi_i^j\left(f_j(x)\right) = f_i(x)$. So $\left(f_i(x)\right)_I \in L$ for all x. We define

$$\begin{aligned} u : X &\longmapsto L \\ x &\longmapsto \left(f_i(x)\right)_I \end{aligned}$$

$\pi_i \circ u(x) = f_i(x)$ and if $\forall u' \neq u, \exists x \mid u(x) \neq u'(x) \implies \exists i \mid f_i(x) \neq \pi_i \circ u(x)$. $\square$

**Definition 1.4.** Let $(M_i)_I$ be a projective system on a well-ordered set, we will say that $(M_i)_I$ satisfies property $(\star)$ when:

- if $i \in I$ has no predecessor then $M_i$ is the limit of $(M_j)_{j \leq i}$,
- if $i$ has a predecessor $i-1$, then $M_i \mapsto M_{i-1}$ is surjective.

**Lemma 1.3.** *Let $I$ be a well-ordered set, and $(M_i)_{i\in I}$, $(N_i)_{i\in I}$, $(P_i)_{i\in I}$ and $(Q_i)_{i\in I}$ be projective systems satisfying $(\star)$. Let*

$$(A_i \overset{f_i}{\longmapsto} B_i \overset{g_i}{\longmapsto} C_i \overset{h_i}{\longmapsto} D_i)_{i\in I}$$

*be morphisms of $A$-modules such that for each $i$, $g_i \circ f_i = 0$ and $h_i \circ g_i = 0$. And let*

$$A \overset{f}{\longmapsto} B \overset{g}{\longmapsto} C \overset{h}{\longmapsto} D$$

*be the limit. If $i$ has a predecessor $i-1$, let $A'_i, B'_i, C'_i, D'_i$ be the respective kernels of the surjections $A_i \mapsto A_{i-1}$, $B_i \mapsto B_{i-1}$, $C_i \mapsto C_{i-1}$ and $D_i \mapsto D_{i-1}$. Let $j \in I$ such that for all $i$ satisfying $j \leq i$, if $i$ has a predecessor then the sequence $A'_i \mapsto B'_i \mapsto C'_i \mapsto D'_i$ is exact. Then the natural morphism*

$$\ker g / im\, f \longmapsto \ker g_j / im f_j$$

*is an isomorphism.*

*Proof.*

$$\begin{tikzcd}
A'_i \arrow[r] \arrow[d, hook] & B'_i \arrow[r] \arrow[d, hook] & C'_i \arrow[r] \arrow[d, hook] & D'_i \arrow[d, hook] \\
A_i \arrow[r, "f_i"] \arrow[d, "{}^A\phi^i"] & B_i \arrow[r, "g_i"] \arrow[d, "{}^B\phi^i"] & C_i \arrow[r, "h_i"] \arrow[d, "{}^C\phi^i"] & D_i \arrow[d, "{}^D\phi^i"] \\
A_{i-1} \arrow[r, "f_{i-1}"] & B_{i-1} \arrow[r, "g_{i-1}"] & C_{i-1} \arrow[r, "h_{i-1}"] & D_{i-1}
\end{tikzcd}$$

$$\begin{aligned}
b \in \ker g_i &\implies g_i(b) = 0 \\
&\implies {}^C\phi^i(g_i(b)) = g_{i-1}({}^B\phi^i(b)) = 0 \\
&\implies {}^B\phi^i(b) \in \ker g_{i-1}
\end{aligned}$$

Let $b_{i-1} \in \ker g_{i-1}$, since ${}^B\phi^i$ is surjective, there exists $b_i \in B_i$ such that ${}^B\phi^i(b_i) = b_{i-1}$ and we have $g_{i-1}(b_{i-1}) = g_{i-1}({}^B\phi^i(b_i)) = {}^C\phi^i(g_i(b_i)) = 0$, so $g_i(b_i) \in \ker {}^C\phi^i = C'_i$. And

$$\begin{aligned}
g_i(b_i) \in \ker h_i = \mathrm{im}\,(g_i : B'_i \mapsto C'_i) &\implies \exists b'_i \in B'_i \mid g_i(b'_i) = g_i(b_i) \\
&\implies g_i(b'_i - b_i) = 0 \\
&\implies b'_i - b_i \in \ker g_i.
\end{aligned}$$

And ${}^B\phi^i(b'_i - b_i) = {}^B\phi^i(b_i) = b_{i-1}$. So for $b_{i-1} \in \ker g_{i-1}$, we found $b'_i - b_i \in \ker g_i$ such that ${}^B\phi^i(b'_i - b_i) = b_{i-1}$. Consequently the morphism $\phi^i : \ker g_i \mapsto \ker g_{i-1}$ is surjective.

Now let the induced morphism $\overline{{}^B\phi^i} : \ker g_i \mapsto \ker g_{i-1}/\operatorname{im} f_{i-1}$. $\forall b \in \ker g_i$ :

$$\begin{aligned}
{}^B\phi^i(b) \in \operatorname{im} f_{i-1} &\implies \exists a_{i-1} \in A_{i-1} \mid f_{i-1}(a_{i-1}) = \phi^i(b) \\
&\implies \exists a \in A_i \mid \phi^i(b) = f_{i-1} \circ \phi^i(a) = \phi^i \circ f_i(a) \\
&\implies \phi^i(b - f_i(a)) = 0 \\
&\implies b - f_i(a) \in B'_j \wedge g_i(b - f_i(a)) = 0 \\
&\implies b - f_i(a) \in B'_j \wedge b - f_i(a) \in \ker g_i \\
&\implies b - f_i(a) \in \operatorname{im} f_i \\
&\implies \exists x \in A'_j \subseteq A_j \mid f_i(x) = b - f_i(a)
\end{aligned}$$

So $b = f_i(x + a) \in \operatorname{im} f_i$, therefore $\ker g_i/\operatorname{im} f_i \simeq \ker g_{i-1}/\operatorname{im} f_{i-1}$.

Next, let $i \neq j$; suppose $i \leq j$.

- If $\phi_i^j = \phi_{j-1}^j \circ \cdots \circ \phi_i^{i+1}$ exists, then it is surjective.
- Otherwise let $k$ be the smallest element of $\left\{k \mid k \leq j \text{ and } \phi_k^j \text{ exists}\right\}$ and so $A_k = \varinjlim (A_l)_{l \leq k}$. Since $i \leq k$ then there exists the natural morphism $\pi : A_k \mapsto A_i$ and $\pi$ is surjective. Indeed it is sufficient to consider the diagram

$$\begin{tikzcd}
 & A_{i+1} \arrow[d, "u"] \arrow[ddl, "\mathrm{id}"'] \arrow[ddr, two heads] & \\
 & A_k \arrow[dl] \arrow[dr] & \\
A_{i+1} \arrow[rr, two heads] & & A_i
\end{tikzcd}$$

So in $I$, $\forall i \leq j$, $\exists \phi : A_j \mapsto A_i$ surjective. Let $\left\{A_i \overset{f_i}{\mapsto} B_i \overset{g_i}{\mapsto} C_i \mid j \leq i\right\}$ ordered by

$$\begin{tikzcd}
A_i \arrow[r] \arrow[d, two heads] & B_i \arrow[r] \arrow[d, two heads] & C_i \arrow[d, two heads] \\
A_i \arrow[r] & B_i \arrow[r] & C_i
\end{tikzcd}$$

. This sequence is totally ordered and bounded above by $A \overset{f}{\mapsto} B \overset{g}{\mapsto} C$. Indeed $\forall i$,

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, two heads] & B \arrow[r, "g"] \arrow[d, two heads] & C \arrow[d, two heads] \\
A_i \arrow[r] & B_i \arrow[r] & C_i
\end{tikzcd}$$

is surjective by the preceding argument. So by Zorn's lemma, there exists $A_m \mapsto B_m \mapsto C_m$ maximal in the sequence. We have: given $(A \xmapsto{\pi_m} A_m)$, if $a \in A$ satisfies $\pi_m(a) = 0$, then $\forall i \in I$; $(\phi : A_m \mapsto A_i)(a) = 0$ so $a = 0$. Consequently

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "\cong"] & B \arrow[r, "g"] \arrow[d, "\cong"] & C \arrow[d, "\cong"] \\
A_m \arrow[r] & B_m \arrow[r] & C_m
\end{tikzcd}$$

is an isomorphism which implies that

$$\ker g / \operatorname{im} f \cong \ker g_m / \operatorname{im} f_m \cong \ker g_j / \operatorname{im} f_j$$

$\square$

## 1.2 Operations on modules

### 1.2.1 Direct sum

**Definition 1.1.** Let $(M_i)_{i \in I}$ be a family of modules. The direct sum of this family of A-modules, denoted $\bigoplus_{i \in I} M_i$, is generally defined in the following two equivalent ways:

- As finite linear combinations :

  The direct sum $\bigoplus_{i \in I} M_i$ is the set of sums of the form

  $$\sum_{i \in I} a_i x_i \quad \text{with} \quad a_i \in A, x_i \in M_i,$$

  of which all terms are zero, except for a finite number of them.

- As a coproduct, that is, if it satisfies the following universal property :

  For any A-module $N$, and any family of morphisms $f_i : M_i \to N$, there exists a unique morphism $f : \bigoplus_{i \in I} M_i \to N$ such that the following diagram is commutative:

$$\begin{tikzcd}
\cdots & M_{i+n} \arrow[dr, "m_{i+n}"] \arrow[ddr, bend right, "f_{i+n}"'] & \cdots & M_i \arrow[d, "m_i"] \arrow[ddl, bend right, "f_i"'] & \cdots & M_{i-n} \arrow[dl, "m_{i-n}"'] \arrow[ddl, bend left, "f_{i-n}"] & \cdots \\
 & & & \bigoplus M_i \arrow[d, dashed, "\exists! f"] & & & \\
 & & & N & & &
\end{tikzcd}$$

that is, for all $i \in I$, $f \circ m_i = f_i$.

*Remark.* When the index set $I$ is finite, then the direct sum is equivalent to the Cartesian product.

### 1.2.2 Tensor product

In this subsection $M$ is a right $A$-module and $N$ a left $A$-module.

**Definition 1.1.** Given $M$ and $N$, let $X$ be an abelian group. A $A$-balanced bilinear map is any abelian group morphism $f : M \times N \mapsto X$ such that, $\forall m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$ and $a \in A$:

1. $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$
2. $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$
3. $f(ma, n) = f(m, an)$

**Definition 1.2.** The tensor product of $M$ by $N$ over the ring $A$ is the abelian group $M \otimes_A N$ which, when equipped with a 'generic' morphism $(\cdot \otimes_A \cdot) : M \times N \mapsto M \otimes_A N$, satisfies the following universal property:

$\forall X$ an abelian group and $f : M \times N \mapsto X$ a $A$-balanced bilinear map, there exists a unique abelian group morphism $u : M \otimes_A N \mapsto X$ such that $u \circ (\cdot \otimes_A \cdot) = f$.

**Proposition 1.1.** *For any right $A$-module $M$, and any left $A$-module $N$, their tensor product $M \otimes_A N$ exists. [2]*

### 1.2.3 Short exact sequence of modules

**Definition 1.1.** A short exact sequence of modules is the data of three modules $M, N, T$, and two module morphisms $f : M \to N, g : N \to T$, satisfying:

- $f$ is injective,
- $g$ is surjective,
- and $\operatorname{im} f = \ker g$.

It is noted as follows: $0 \to M \xrightarrow{f} N \xrightarrow{g} T \to 0$.

*Remark.* Any short exact sequence $0 \to M \xrightarrow{f} N \xrightarrow{g} T \to 0$ satisfies: $T \cong N/\operatorname{im} f$.

**Definition 1.2.** Let the short exact sequence $0 \to M \xrightarrow{f} N \xrightarrow{g} T \to 0$, then we say that it is split if there exists an application $g' : T \to N$ such that $g \circ g' = 1_T$. The following lemma follows.

**Lemma 1.1** (Splitting lemma)**.** *For the split short exact sequence of the form: $0 \to M \xrightarrow{f} N \xrightarrow{g} T \to 0$, we have the relation $N \cong M \oplus T$.*

*Proof.* See reference [3]. □

**Proposition 1.2.** *For any A-module $X$, and any short exact sequence $0 \to M \xrightarrow{f} N \xrightarrow{g} T \to 0$, the sequence:*

$$X \otimes_A M \xrightarrow{1_X \otimes_A f} X \otimes_A N \xrightarrow{1_X \otimes_A g} X \otimes_A T \to 0$$

*is exact.*

## 1.3 Some types of modules

**Definition 1.1.** Let $M$ be a left A-module, we say that $M$ is finitely generated if there exists a finite subset $B$ of $M$ such that every element of $M$ can be written as a linear combination of elements of $B$. That is, $\forall x \in M$, there exists a sequence $(a_b)_{b \in B} \subset A$ such that:

$$x = \sum_{b \in B} a_b b$$

**Proposition 1.1.** *Any quotient of a finitely generated module is also finitely generated.*

*Proof.* Let $N$ be a submodule of $M$, and $x \in M/N$. We have,

$$x = \sum_{b \in B} a_b b = \sum_{b \in B} a_b \bar{b}$$

Hence the finite set $\bar{B} = \{\bar{b} | b \in B\}$, generates $M/N$. □

**Definition 1.2.** Let $M$ be an A-module. We say that $M$ is a Noetherian module, if one of the following equivalent conditions is satisfied:

- Every totally ordered set of submodules of $M$ has a maximal element.
- Every increasing sequence of submodules of $M$ is stationary.
- Every submodule of $M$ is finitely generated.

**Definition 1.3** (Noetherian Ring)**.** A ring $A$ is said to be Noetherian, if it is Noetherian as an A-module.

*Remark.* Every Noetherian module is finitely generated. But the converse is false.

**Proposition 1.2.** *If $A$ is a Noetherian ring, then for any A-module $M$, we have:*

$$M \text{ is Noetherian} \iff M \text{ is finitely generated}$$

*Proof.* See reference [2]. □

### 1.3.1 Free module

**Definition 1.1.** Let $M$ be a left A-module. We say that the module $M$ is free if there exists a subset $B$ of $M$ such that, every element $x$ of $M$ can be written in a unique way as a linear combination of elements of $B$. Equivalently, we will say that $M$ is free if there exists $B \subset M$ a subset such that:

$$\sum_{b \in B} a_b b = 0 \implies a_b = 0, \forall b \in B,$$

with the $a_b$ being elements of $A$. And in this case, we call $B$, the basis of the module $M$.

**Proposition 1.1.** *Every module is a quotient of a free module.*

*Proof.* Let $M$ be an A-module, and $F$ the free A-module with basis $B = \{e_m | m \in M\}$. Let the morphism $f : F \to M$, such that $f(e_m) = m$. Then $f$ is surjective and we have $M \cong F/\ker f$. □

### 1.3.2 Projective module

**Definition 1.1.** An A-module $P$ is said to be projective if, for any surjective morphism $f : M \to N$ between two A-modules, and any morphism $\phi : P \to N$, there exists a unique morphism $g : P \to M$ such that $f \circ g = \phi$. In other words, any diagram with an exact row of this form

$$\begin{tikzcd} & P \arrow[d, "\phi"] & \\ M \arrow[r, "f"] & N \arrow[r] & 0 \end{tikzcd}$$

can be completed by a unique morphism $g$ to give the following commutative diagram:

$$\begin{tikzcd} & P \arrow[d, "\phi"] \arrow[dl, dashed, "\exists g"'] & \\ N \arrow[r, "f"] & N \arrow[r] & 0 \end{tikzcd}$$

**Proposition 1.1.** *If the A-module $P$ is projective, then any exact sequence of A-modules of the form $0 \to N \to M \xrightarrow{p} P \to 0$ is split.*

*Proof.* Since $P$ is projective, there exists then $\phi : P \to M$ such that $f \circ \phi = \mathrm{id}_P$, hence the result. □

**Proposition 1.2.** *Every free module is projective.[2]*

### 1.3.3 Injective modules

**Definition 1.1.** Let $I$ be an A-module, then we say that $I$ is injective if for any injective morphism $f' : M' \to N'$ between two A-modules, and for any morphism $\psi : M' \to I$, there exists a unique morphism $g' : N' \to I$ such that $g' \circ f' = \psi$. In other words, any diagram of this form

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "f'"] \arrow[d, "\psi"] & N' \\ & I & \end{tikzcd}$$

with an exact row, can be completed, to give the following commutative diagram:

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "f'"] \arrow[d, "\psi"] & N' \arrow[dl, dashed, "\exists g"] \\ & I & \end{tikzcd}$$

**Proposition 1.1.** *If the A-module $I$ is injective, then any exact sequence of A-modules of the form $0 \to I \xrightarrow{i} M' \to N' \to 0$ is split.*

*Proof.* Since $I$ is injective, there exists a morphism $i' : M' \to I$ such that $i' \circ i = 1_I$. Hence the sequence is split. □

**Theorem 1.2.** *Every A-module is a submodule of an injective A-module called its injective envelope.*

### 1.3.4 Flat module

**Definition 1.1.** An A-module $P$ is said to be flat, if for any injective morphism of A-modules $f : M \to N$, we have: $M \otimes_A P \to N \otimes_A P$ is also an injective morphism.

**Proposition 1.1.** *Every projective module is flat, by transitivity, every free module is also flat. [2]*

**Proposition 1.2.** *For any flat A-module $F$, and any short exact sequence $0 \to M \xrightarrow{f} N \xrightarrow{g} T \to 0$, the sequence*

$$0 \to F \otimes_A M \xrightarrow{1_F \otimes_A f} F \otimes_A N \xrightarrow{1_F \otimes_A g} F \otimes_A T \to 0$$

*is also exact.*

## 1.4 Bimodule

**Definition 1.1.** Let $A$, and $B$, be two rings, we say that $M$ is an A-B-bimodule, if $M$ can be considered as a left A-module, but also as a right B-module, such that the following condition is satisfied for all $x \in M, a \in A$, and $b \in B$ :

$$(ax)b = a(xb)$$

**Proposition 1.1.** *Let $A, B$ and $C$ be three rings, $M$ is a B-A-bimodule, and $N$ is an A-C-bimodule, then the tensor product $M \otimes_A N$ is a B-C-bimodule.*

# Chapter 2

# The category of complexes of $A$-modules : $\mathrm{Comp}\,(A\text{-Mod})$

Homological algebra is strongly connected with algebraic topology which studies the classification of topological spaces up to homeomorphism. In homological algebra, the homology groups of a space are the main object of interest, because they define in the case of topological spaces invariants preserved by homeomorphisms. To define and study these homology groups, we will introduce the notion of category, which is the basic structure on which we will work, but above all, the homology functor defined on the category of complex sequences on an abelian category (here it will be the category of left or right $A$-modules).

## 2.1 Category and functors

### 2.1.1 Category

**Definition 2.1.** We call category, let's note it $\mathcal{C}$, a structure consisting of:

1. A class of objects, noted: $\mathrm{ob}\,(\mathcal{C})$.
2. For any pair of objects $(X, Y)$ in $\mathrm{ob}\,(\mathcal{C})$, there is associated a set of morphisms of source $X$ and target $Y$, noted $\mathrm{Hom}_{\mathcal{C}}(X, Y)$. These morphisms are sometimes also called arrows, since these are axiomatic objects of category theory, and not morphisms in the sense of applications in set theory.
3. For any $X, Y, Z \in \mathrm{ob}\,(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(Y, Z)$; there exists a morphism $g \circ f \in \mathrm{Hom}(X, Z)$ called the composite morphism of $f$ and $g$.
4. For any $X \in \mathrm{ob}\,(\mathcal{C})$, we have a morphism $1_X \in \mathrm{Hom}_{\mathcal{C}}(X, X)$, called the identity morphism, such that $\forall Y \in \mathrm{ob}\,(\mathcal{C})$, we have:
   (a) $f \circ 1_X = f, \forall f \in \mathrm{Hom}_C(X, Y)$

(b) $1_X \circ g = g, \forall g \in \mathrm{Hom}_C(Y, X)$

**Example.** $A$-Mod: the category whose objects are left $A$-modules, and the morphisms are left $A$-module morphisms.

**Definition 2.2** (Opposite Category)**.** Let $\mathcal{C}$ be a category. The opposite category to the category $\mathcal{C}$, noted $\mathcal{C}^{\mathrm{op}}$, is the category whose objects are the objects of $\mathcal{C}$, and the morphisms are the opposite morphisms to those of $\mathcal{C}$. That is, if $f : X \to Y$ is a morphism of $\mathcal{C}$, then it corresponds to the opposite arrow $f : Y \to X$ in $\mathcal{C}^{op}$.

### 2.1.2 Functor

**Definition 2.1.** For $A$ and $B$ categories, we can define a morphism between these two categories. It is sufficient for this morphism to respect the internal structure of these categories. We call this type of correspondence between categories a covariant (respectively, contravariant) functor, and it is defined as follows: Let $F : A \to B$ satisfying the following axioms:

1. $\forall X \in \mathrm{ob}\,(A), F(X) \in \mathrm{ob}(B)$
2. $\forall f \in \mathrm{Hom}_A(X, Y)$, we have $F(f) \in \mathrm{Hom}_B(F(X), F(Y))$ (respectively, $F(f) \in \mathrm{Hom}_B(F(Y), F(X))$) satisfying:
   (a) $\forall f \in \mathrm{Hom}_A(X, Y), g \in \mathrm{Hom}_A(Y, Z), F(g \circ f) = F(g) \circ F(f)$ (respectively, $F(g \circ f) = F(f) \circ F(g)$).
   (b) $\forall X \in \mathrm{ob}(C); F(1_X) = 1_{F(X)}$

*Remark.* It is clear that in the case of a contravariant functor, the arrows between the objects of the starting category are reversed between the images of these objects in the arrival category. Thus defining a contravariant functor F from a category A to a category B is equivalent to defining a covariant functor from the category A to the category $B^{op}$, or equivalently from the category $A^{op}$ to B. That is: $F : A \to B$ contravariant $\iff$ $F : A \to B^{op}$ is covariant $\iff$ $F : A^{op} \to B$ is covariant.

### 2.1.3 Natural Transformations

**Definition 2.1.** Between two functors, of the same source, and of the same target, one can define a morphism, which preserves the functorial structure. This type of morphism between functors is called a natural transformation, and is defined as follows: Let A and B be two categories, $F : A \to B$ and $G : A \to B$ be two covariant (respectively, contravariant) functors. Then a natural transformation $\alpha$ from the functor A to the functor B consists of the following data:

1. For any $X \in \mathrm{ob}(A)$, we associate a morphism $\alpha_X : F(X) \to G(X)$ in B.
2. For any morphism $f : X \to Y$ in A, the following diagram is commutative.

$$\begin{tikzcd} F(X) \arrow[r, "F(f)"] \arrow[d, "\alpha_X"] & F(Y) \arrow[d, "\alpha_Y"] \\ G(X) \arrow[r, "G(f)"] & G(Y) \end{tikzcd}$$

(Respectively, for the contravariant case:

$$\begin{tikzcd} F(Y) \arrow[r, "F(f)"] \arrow[d, "\alpha_Y"] & F(X) \arrow[d, "\alpha_X"] \\ G(Y) \arrow[r, "G(f)"] & G(X) \end{tikzcd}$$

)

Such a morphism between functors is often noted $F \overset{\alpha}{\Longrightarrow} G$.

And when for any object X of A, the morphism $\alpha_X$ is an isomorphism, we say that $\alpha$ is a natural isomorphism.

## 2.2 Category of complexes of A-module

### 2.2.1 Complexes

**Definition 2.1.** A complex of $A$-module is an object consisting of a sequence of $A$-modules $(M_n)_{n\in\mathbb{Z}}$, linked together by morphisms $(\delta_n : M_n \to M_{n-1})_{n\in\mathbb{Z}}$, such that for all $n \in \mathbb{Z}$,

$$\delta_n \circ \delta_{n+1} = 0,$$

so $\operatorname{im}(\delta_{n+1}) \subseteq \ker(\delta_n)$. It is often represented in the following way:

$$\dots \xrightarrow{\delta_{n+2}} M_{n+1} \xrightarrow{\delta_{n+1}} M_n \xrightarrow{\delta_n} M_{n-1} \xrightarrow{\delta_{n-1}} \dots$$

With this notation, $\delta_n$ is sometimes called the differential of degree $n$. When we have the equality $\operatorname{im}(\delta_{n+1}) = \ker(\delta_n), \forall n \in \mathbb{Z}$ we say that this sequence is exact. For any $A$-module $N$, we will also denote by $N$ the complex of $A$-module which is the $A$-module $N$ at index 0, and is the $A$-module 0 everywhere else:

$$\dots \to 0 \to N \to 0 \to \dots$$

*Notations.* For a complex $M_\bullet$:

- We say that $M_\bullet$ is bounded below if there exists $b \in \mathbb{Z}$ such that $M_i = 0$, $\forall i < b$,

$$\dots \mapsto M_{b+3} \mapsto M_{b+2} \mapsto M_{b+1} \mapsto M_b \mapsto 0 \mapsto 0 \mapsto \dots$$

- We say that $M_\bullet$ is bounded above if there exists $b \in \mathbb{Z}$ such that $M_i = 0$, $\forall i > b$,

$$\cdots \mapsto 0 \mapsto 0 \mapsto M_b \mapsto M_{b-1} \mapsto M_{b-2} \mapsto M_{b-3} \mapsto \cdots$$

**Definition 2.2.** Given a complex $M_\bullet$, let $i \in \mathbb{Z}$, the complex $M[i]_\bullet$ is defined by:

- $M[i]_j = M_{i+j}$,
- $d_j^{M[i]} = (-1)^i d_{j+i}^M$.

**Definition 2.3.** Let $M_\bullet$ and $N_\bullet$ be two complexes of $A$-module, we define a morphism of complexes of A-modules from $M_\bullet$ to $N_\bullet$, called a chain complex of $A$-module morphisms, as being a sequence of $A$-module morphisms $(f_n : M_n \to N_n)_{n\in\mathbb{Z}}$ such that $\forall n \in \mathbb{Z}$:

$$f_{n-1} \circ \delta_n = \delta_n \circ f_n$$

That is to say that the following diagram is commutative:

$$\begin{tikzcd}
\dots \arrow[r, "\delta_{n+2}"] & M_{n+1} \arrow[r, "\delta_{n+1}"] \arrow[d, "f_{n+1}"] & M_n \arrow[r, "\delta_n"] \arrow[d, "f_n"] & M_{n-1} \arrow[r, "\delta_{n+1}"] \arrow[d, "f_{n-1}"] & \dots \\
\dots \arrow[r, "\delta_{n+2}"] & N_{n+1} \arrow[r, "\delta_{n+1}"] & N_n \arrow[r, "\delta_n"] & N_{n-1} \arrow[r, "\delta_{n+1}"] & \dots
\end{tikzcd}$$

By defining what a complex of $A$-module is, and what a morphism of complexes of $A$-modules is, we have given what the category of A-module complexes, denoted $\mathrm{Comp}\,(A\text{-Mod})$, consists of.

Let us verify that it forms a category.

*Proof.*

1. The complexes of $A$-modules form the class of objects $\mathrm{ob}\,(\mathrm{Comp}\,(A\text{-Mod}))$.
2. For all $M_\bullet, N_\bullet \in \mathrm{ob}\,(\mathrm{Comp}\,(A\text{-Mod}))$, $\mathrm{Hom}_{\mathrm{Comp}\,(A\text{-Mod})}(M_\bullet, N_\bullet)$ is the set of morphisms of complexes of $A$-modules from $M_\bullet$ to $N_\bullet$.
3. Let $M_\bullet, N_\bullet, T_\bullet$ be complexes of $A$-modules, and let $f : M_\bullet \to N_\bullet$, $g : N_\bullet \to T_\bullet$ be morphisms of complexes of $A$-modules. Then the composite $g \circ f$ defined by
$$(g \circ f)_i = g_i \circ f_i, \quad \forall i \in \mathbb{Z},$$

is a morphisms of complexes of $A$-modules from $M_\bullet$ to $T_\bullet$, since the following diagram is commutative:

$$\begin{tikzcd}
\dots \arrow[r] & M_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "f_{i+1}"] & M_i \arrow[r, "d_i"] \arrow[d, "f_i"] & M_{i-1} \arrow[r] \arrow[d, "f_{i-1}"] & \dots \\
\dots \arrow[r] & N_{i+1} \arrow[r, "d'_{i+1}"] \arrow[d, "g_{i+1}"] & N_i \arrow[r, "d'_i"] \arrow[d, "g_i"] & N_{i-1} \arrow[r] \arrow[d, "g_{i-1}"] & \dots \\
\dots \arrow[r] & T_{i+1} \arrow[r, "d''_{i+1}"] & T_i \arrow[r, "d''_i"] & T_{i-1} \arrow[r] & \dots
\end{tikzcd}$$

4. Let $M_\bullet$ be a complex sequence of morphisms of $A$-modules. Define $1_{M_\bullet} = (1_{M_i})_{i\in\mathbb{Z}}$. Then for any $N_\bullet \in \mathrm{ob}\,(\mathrm{Comp}\,(A\text{-Mod}))$ and any $f : M_\bullet \to N_\bullet$,

$$f \circ 1_{M_\bullet} = (f_i \circ 1_{M_i})_{i\in\mathbb{Z}} = f,$$

and for any $g : N_\bullet \to M_\bullet$,

$$1_{M_\bullet} \circ g = (1_{M_i} \circ g_i)_{i\in\mathbb{Z}} = g.$$

□

**Definition 2.4.** Let $M_\bullet, N_\bullet \in \mathrm{ob}\,(\mathrm{Comp}\,A\text{-Mod})$, and $f : M_\bullet \to N_\bullet$ a morphism of complexes of $A$-modules. Then we say that the morphism $f$ is injective if for all $n \in \mathbb{Z}$, $f_n : M_n \to N_n$ is injective.

Similarly, to say that $f$ is surjective, is to say that $f_n$ is surjective for all $n \in \mathbb{Z}$.

**Definition 2.5.** For any morphism of complexes of $A$-modules $f : M_\bullet \to N_\bullet$, the kernel of the morphism $f$ is the complex of $A$-modules noted $\ker f$, given by, $\forall n \in \mathbb{Z}$:

1. $(\ker f)_n = \ker(f_n)$,
2. and the differential is the restriction of the differential of the complex of $A$-modules $M_\bullet$ on $\ker f$, that is:

$$\begin{aligned} \delta_n : \ker f_n &\mapsto \ker f_{n-1} \\ x &\mapsto \delta_n(x) \end{aligned}$$

with $\delta_n : M_n \to M_{n-1}$, the differential of degree $n$ of the complex of $A$-modules $M_\bullet$.

**Definition 2.6.** For any morphism of complexes of $A$-modules $f : M_\bullet \to N_\bullet$, as for the kernel, we define the image of the morphism $f$, noted $\mathrm{im}\, f$, as the complex of $A$-modules

$$\cdots \to \mathrm{im}\, f_{n+1} \xrightarrow{d_{n+1}} \mathrm{im}\, f_n \xrightarrow{d_n} \mathrm{im}\, f_{n-1} \to \dots$$

*Remark.* As for modules, a short exact sequence of complexes of $A$-modules, not to be confused with a short exact sequence of $A$-module, consists of the following data :

Let $M_\bullet, N_\bullet, T_\bullet \in \operatorname{ob}(\operatorname{Comp} A\text{-Mod})$, $f \in \operatorname{Hom}_{\operatorname{Comp} A\text{-Mod}}(M_\bullet, N_\bullet)$, and $g \in \operatorname{Hom}_{\operatorname{Comp} A\text{-Mod}}(N_\bullet, T_\bullet)$ be chain morphisms of complexes $A$-modules, then if $f$ is injective, $g$ surjective, and $\ker f = \operatorname{im} g$, we say that the sequence :

$$0 \to M_\bullet \to N_\bullet \to T_\bullet \to 0$$

is a short exact sequence of complexes of $A$-modules.

### 2.2.2 Homotopic chain complexes of A-module

**Definition 2.1.** Let $M_\bullet$ and $N_\bullet$ be two complexes of $A$-modules, and $f_\bullet$ and $g_\bullet$ two morphisms of complexes from $M_\bullet$ to $N_\bullet$. A homotopy between $f_\bullet$ and $g_\bullet$ is a sequence of $A$-module morphisms $(s_n : M_n \mapsto N_{n+1})_{n\in\mathbb{Z}}$ such that, $\forall n \in \mathbb{Z}$, $f_n - g_n = s_{n-1} \circ d_n^M + d_{n+1}^N \circ s_n$. We denote it by $s : f_\bullet \sim g_\bullet$.

$$\begin{tikzcd}
\dots \arrow[r, "d_{i+2}"] & M_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, shift right, "g_{i+1}"'] \arrow[d, shift left, "f_{i+1}"] & M_i \arrow[r, "d_i"] \arrow[d, shift right, "g_i"'] \arrow[d, shift left, "f_i"] \arrow[dl, "s_i"] & M_{i-1} \arrow[r, "d_{i-1}"] \arrow[d, shift right, "g_{i-1}"'] \arrow[d, shift left, "f_{i-1}"] \arrow[dl, "s_{i-1}"] & \dots \\
\dots \arrow[r, "d_{i+2}"] & N_{i+1} \arrow[r, "d_{i+i}"] & N_i \arrow[r, "d_i"] & N_{i-1} \arrow[r, "d_{i-1}"] & \dots
\end{tikzcd}$$

When $f$ is homotopic to the null chain complex of morphisms, we say that $f$ is null-homotopic.

**Definition 2.2.** Let $M \in \operatorname{ob}(\text{comp A-Mod})$, if $1_M$ is null-homotopic, we say that $M$ is a contractible complex of A-module morphisms.

**Proposition 2.1.** *The homotopy relation induces an equivalence relation.*

*Proof.* We have:

- $0 : f_\bullet \sim f_\bullet$.
- $s : f_\bullet \sim g_\bullet$, and $s' : g_\bullet \sim h_\bullet$, so $\forall n \in \mathbb{Z}$, $f_n - g_n = s_{n-1} \circ d_n^M + d_{n+1}^N \circ s_n$ and $g_n - h_n = s'_{n-1} \circ d_n^M + d_{n+1}^N \circ s'_n$. So $\forall n \in \mathbb{Z}$,

$$\begin{aligned}
h_n - f_n &= h_n - g_n + g_n - f_n \\
&= s_{n-1} \circ d_n^M + d_{n+1}^N \circ s_n + s'_{n-1} \circ d_n^M + d_{n+1}^N \circ s'_n \\
&= (s_{n-1} + s'_{n-1}) \circ d_n^M + d_{n+1}^N \circ (s_n + s'_n)
\end{aligned}$$

  so $s + s' : f_\bullet \sim h_\bullet$
- $s : f_\bullet \sim g_\bullet \implies \forall n \in \mathbb{Z},\ f_n - g_n = -s_{n-1} \circ d_n^M + d_{n+1}^N \circ (-s_n)$, so $(-s) : g_\bullet \sim f_\bullet$.

□

**Definition 2.3.** Let $M_\bullet$ and $N_\bullet$ be two complexes of $A$-modules. We say that $M_\bullet$ and $N_\bullet$ are homotopic if there exist two morphisms $f_\bullet : M_\bullet \mapsto N_\bullet$ and $g_\bullet : N_\bullet \mapsto M_\bullet$ such that $f_\bullet \circ g_\bullet \sim 1_{N_\bullet}$ and $g_\bullet \circ f_\bullet \sim 1_{M_\bullet}$.

**Proposition 2.2.** *Let the two homotopies $s : f_\bullet \sim g_\bullet : M_\bullet \mapsto M'_\bullet$ and $s' : f'_\bullet \sim g'_\bullet : M'_\bullet \mapsto M''_\bullet$. Then we have the homotopy $f' \circ s + s' \circ g : f \circ f' \sim g \circ g' : M \mapsto M''$.*

### 2.2.3 Homology of a complex of A-module

**Definition 2.1.** For any complex of A-module morphisms M, we call cycle of degree n, the submodule of the A-module $M_n$ determined by the kernel of the differential of degree n (that is, $\delta_n : M_n \to M_{n-1}$). It is noted $Z_n(M)$.

$$Z_n(M) = \ker \delta_n$$

**Definition 2.2.** For any complex of A-module morphisms M, we call boundary of degree n, the submodule of the A-module $M_n$ determined by the image of the differential of degree $n+1$. It is noted $B_n(M)$.

$$B_n(M) = \operatorname{im} \delta_{n+1}$$

*Remark.* For all $n \in \mathbb{Z}$, we have $B_n(M) \subseteq Z_n(M)$.

**Definition 2.3.** Let M be a complex of A-module morphisms, then $\forall n \in \mathbb{Z}$, the homology of degree n of the complex of A-module morphisms M is the A-module defined as follows:

$$H(M)_n = Z_n(M)/B_n(M)$$

For each index n, we speak of the homology functor of degree $n$ :

$$\begin{aligned} H_n : \text{comp(A-Mod)} &\mapsto \text{A-Mod} \\ \text{ob(comp A-Mod)} \ni M &\mapsto H_n(M) \\ \text{Hom}_{\text{comp A-Mod}}(M, N) \ni f &\mapsto (H_n(f) : H_n(M) \to H_n(N)) \end{aligned}$$

**Definition 2.4.** Let $M$ be a complex of $A$-modules,

- We say that $M$ is acyclic, when $H_i(M) = 0$, $\forall i \in \mathbb{Z}$
- $\sup(M_\bullet) = \sup\{i \in \mathbb{Z} \mid H_i(M_\bullet) \neq 0\}$
- $\inf(M_\bullet) = \inf\{i \in \mathbb{Z} \mid H_i(M_\bullet) \neq 0\}$

**Proposition 2.1.** $H_j(M[i]) = H_{j+i}(M)$.

**Definition 2.5** (Quasi-isomorphism)**.** Let $M, N \in$ ob(comp A-Mod), and $f : M \to N$ a chain complex of A-module morphisms, if for all $n \in \mathbb{Z}, H_n(f) : H_n(M) \to H_n(N)$ is an isomorphism, we say that $f$ is a quasi-isomorphism (or homology isomorphism).

**Proposition 2.2.** *Any short exact sequence of complexes* $0 \mapsto M_\bullet^{'} \overset{f_\bullet}{\longmapsto} M_\bullet \overset{g_\bullet}{\longmapsto} M_\bullet^{''} \mapsto 0$*, induces a long exact sequence:*

$$\cdots \overset{H_{i+1}(g_\bullet)}{\longmapsto} H_{i+1}(M_\bullet^{''}) \overset{\partial_{i+1}}{\longmapsto} H_i(M_\bullet^{'}) \overset{H_i(f_\bullet)}{\longmapsto} H_i(M_\bullet) \overset{H_i(g_\bullet)}{\longmapsto} M_i^{''} \overset{\partial_i}{\longmapsto} H_{i-1}(M_\bullet^{'}) \overset{H_{i-1}(f_\bullet)}{\longmapsto} \cdots$$

*Proof.* Let's proceed with the following diagram:

$$\begin{tikzcd}
\dots \arrow[r] & M_{n+1} \arrow[r, "d_{n+1}"] \arrow[d, "f_{n+1}"] & M_n \arrow[r, "d_n"] \arrow[d, "f_n"] & M_{n-1} \arrow[r] \arrow[d, "f_{n-1}"] & \dots \\
\dots \arrow[r] & N_{n+1} \arrow[r, "d_{n+1}"] \arrow[d, "g_{n+1}"] & N_n \arrow[r, "d_n"] \arrow[d, "g_n"] & N_{n-1} \arrow[r] \arrow[d, "g_{n-1}"] & \dots \\
\dots \arrow[r] & T_{n+1} \arrow[r, "d_{n+1}"] & T_n \arrow[r, "d_n"] & T_{n-1} \arrow[r] & \dots
\end{tikzcd}$$

Let the morphism be:

$$\begin{aligned}
\sigma_n : H_n(T) &\mapsto H_{n-1}(M) \\
\overline{t_n} &\mapsto \overline{f_{n-1}^{-1} \circ d_n \circ g_n^{-1}(\{t_n\})}
\end{aligned}$$

- Let's show this morphism is well-defined.

  First, since $g_n$ is surjective, every element of $T_n$ has a preimage, in particular every element of $Z_n(T)$. Let $A = g_n^{-1}(\{t\})$, then $g_{n-1} \circ d_n(A) = d_n \circ g_n(A) = 0$, so $d_n(A) \subseteq \ker g_{n-1} = \operatorname{im} f_{n-1}$, which implies $f_{n-1}^{-1}(d_n(A))$ is defined.

  Also let $t, t_0 \in Z_n(T)$ such that $t - t_0 \in B_n(T)$. Let's check that $\sigma_n(t) = \sigma_n(t_0)$:

  $$t - t' \in B_n(T) \implies \exists X \subseteq T_{n+1} \text{ such that } d_{n+1}(X) = \{t - t'\},$$

  and since $g_{n+1}$ is surjective, there exists $A \subseteq N_{n+1}$ such that $g_{n+1}(A) = X$. Since g is a chain map of $A$-modules, we have $g_n \circ d_{n+1} = d_{n+1} \circ g_{n+1}$, so

  $$g_n \circ d_{n+1}(A) = d_{n+1} \circ g_{n+1}(A) \implies g_n(d_{n+1}(A)) = \{t - t'\}$$

  Let $B = d_{n+1}(A)$, that is, $g_n^{-1}(\{t - t'\}) = B$. And if we evaluate $d_n(B)$, we find $\{0\}$, because $B = d_{n+1}(A)$. So since $f_{n-1}$ is injective, we conclude that if $\overline{t} = \overline{t'}$ then $\sigma_n(\overline{t}) = \sigma_n(\overline{t'})$, hence $\sigma_n$ is well-defined.

- Let's show the sequence:

  $$H_n(N) \xrightarrow{H_n(g)} H_n(T) \xrightarrow{\sigma_n} H_{n-1}(M) \xrightarrow{H_{n-1}(f)} H_{n-1}(N)$$

  is exact.

– Let's show $\operatorname{im} H_n(g) = \ker \sigma_n$.

$$\begin{aligned}
\bar{t} \in \operatorname{im} H_n(g) &\iff \bar{t} \in H_n(g)(H_n(N)) \\
\bar{t} \in \operatorname{im} H_n(g) &\iff \bar{t} \in H_n(g)(Z_n(N)/B_n(N)) \\
&\iff t \in g(Z_n(N)) \\
&\implies \exists x \in Z_n(N) \text{ such that } g_n(x) = t,
\end{aligned}$$

and thus $d_n(x) = 0$. Since $\sigma_n$ is well-defined, and does not depend on the representative in $g_n^{-1}(\{t\})$, then:

$$\sigma_n(\bar{t}) = \overline{f_{n-1}^{-1} \circ d_n(x)} = 0$$

Hence $\bar{t} \in \ker \sigma_n$, and so

$$\operatorname{im} H_n(g) \subseteq \ker \sigma_n \tag{2.1}$$

$$\begin{aligned}
\bar{t} \in \ker \sigma_n &\iff \sigma_n(\bar{t}) = \overline{f_{n-1}^{-1} \circ d_n \circ g_n^{-1}(\{t\})} = 0 \\
&\iff f_{n-1}^{-1} \circ d_n \circ g_n^{-1}(\{t\}) \subseteq B_{n-1}(M) \\
&\iff d_n \circ g_n^{-1}(\{t\}) \subseteq f_{n-1}(B_{n-1}(M)) \\
&\iff d_n \circ g_n^{-1}(\{t\}) \subseteq f_{n-1} \circ d_n(M_n) = d_n \circ f_n(M_n)
\end{aligned}$$

So there exist $a, b \in g_n^{-1}(\{t\})$ such that $d_n(a) \in f_{n-1} \circ d_n(M_n)$, $d_n(b) \in d_n \circ f_n(M_n)$ and $d_n(a) = d_n(b)$. Consequently $a - b \in Z_n(N)$, and

$$\begin{aligned}
g_n(a-b) &= g_n(a) - g_n(b) \\
&= g_n(a) - 0 \\
&= t
\end{aligned}$$

Since $\sigma_n$ is well-defined, and does not depend on the representative in $g_n^{-1}(\{t\})$, then $\bar{t} \in \operatorname{im} H_n(g)$, consequently:

$$\ker \sigma_n \subseteq \operatorname{im} H_n(g). \tag{2.2}$$

2.1 and 2.2 imply that $\ker \sigma_n = \operatorname{im} H_n(g)$.

– Let's show $\operatorname{im} \sigma_n = \ker H_{n-1}(f)$.

$$\begin{aligned}
\overline{m} \in \operatorname{im} \sigma_n &\iff \overline{m} \in \sigma_n(H_n(T)) \\
&\iff \overline{m} \in \overline{f_{n-1}^{-1} \circ d_n \circ g_n^{-1}(Z_n(T))} \\
&\iff m \in f_{n-1}^{-1} \circ d_n \circ g_n^{-1}(Z_n(T)) \\
&\iff f_{n-1}(m) \in d_n \circ g_n^{-1}(Z_n(T)) \\
\overline{m} \in \ker H_{n-1}(f) &\iff H_{n-1}(f)(\overline{m}) = \overline{f_{n-1}(m)} = 0 \\
&\iff f_{n-1}(m) \in B_{n-1}(N) = d_n(N_n) \\
&\iff f_{n-1}(m) \in d_n(N_n) \cap \operatorname{im} f_{n-1}
\end{aligned}$$

Let $x \in g_n^{-1}(Z_n(T))$

$$\begin{aligned} x \in g_n^{-1}(Z_n(T)) &\iff g_n(x) \in Z_n(T) \\ &\iff d_n \circ g_n(x) = 0 \\ &\iff g_{n-1} \circ d_n(x) = 0 \\ &\iff d_n(x) \in \ker g_{n-1} \\ &\iff d_n(x) \in \operatorname{im} f_{n-1} \end{aligned}$$

To conclude, consider the following equivalences:

$$\begin{aligned} x \in d_n \circ g_n^{-1}(Z_n(T)) &\iff \exists y \in g_n^{-1}(Z_n(T)) \text{ such that } d_n(y) = x \\ &\iff d_n(y) \in \operatorname{im} f_{n-1} \text{ and } d_n(y) = x \\ &\iff x \in d_n(N_n) \cap \operatorname{im} f_{n-1} \end{aligned}$$

□

**Proposition 2.3.** *Let $M_\bullet, M_\bullet \in \mathcal{C}(A\text{-}Mod)$ and $f_\bullet, g_\bullet \in \hom(M_\bullet, N_\bullet)$ be two homotopic complex morphisms. Then $\forall n \in \mathbb{Z}$, $H_n(f) = H_n(g)$.*

*Proof.* Since $f \sim g$, there exists $(s_n)_{n\in\mathbb{Z}}$ such that $\forall n \in \mathbb{Z}$, $f_n - g_n = s_{n-1} \circ d_n^M + d_{n+1}^N \circ s_n$. Let $m_n \in Z_n(M)$, then

$$\begin{aligned} (f_n - g_n)(m_n) &= s_{n-1} \circ d_n^M(m_n) + d_{n+1}^N \circ s_n(m_n) \\ &= d_{n+1}^N \circ s_n(m_n) \in B_n(N) \end{aligned}$$

so $\forall \overline{m_n} \in H_n(M)$, $H_n(f)(\overline{m_n}) - H_n(g)(\overline{m_n}) = H_n(f-g)(\overline{m_n}) = \overline{0}$, consequently $H_n(f)(\overline{m_n}) = H_n(g)(\overline{m_n})$, so $H_n(f) = H_n(g)$. □

### 2.2.4 The cone

**Definition 2.1.** Given the morphism of complexes $f_\bullet : M_\bullet \mapsto N_\bullet$, the cone of $f_\bullet$ denoted $\operatorname{cone}(f_\bullet)$ is the complex defined by:

- $(\operatorname{cone} f_\bullet)_i = M_{i-1} \oplus N_i$,
- $d_i(m, n) = (-d_{i-1}^M(m), d_i^N(n) - f_{i-1}(m))$.

**Proposition 2.1.** *The cone of $f : M_\bullet \mapsto N_\bullet$ induces the exact sequence*

$$0 \mapsto N_\bullet \overset{x}{\mapsto} \operatorname{cone} f_\bullet \overset{\pi}{\mapsto} M_\bullet[-1] \mapsto 0$$

*with $x_i(n) = (0, n)$ and $\pi_i(m, n) = -m$.*

*Proof.* 1. We have

- $d_i^{\text{cone}} \circ x_i(n) = d_i^{\text{cone}}(0, n) = (0, d_i^N(n))$,
- $x_{i-1} \circ d_i^N = (0, d_i^N(n))$.

So $x$ is a morphism of complexes.

2. We have that $d_i^{M[-1]} = -d_{i-1}^M$ and $M[-1]_i = M_{i-1}$. And for $(m,n) \in (\operatorname{cone} f)_i = M_{i-1} \oplus N_i$:

   - $d_i^{M[-1]} \circ \pi_i(m,n) = -d_{i-1}^M(-m) = d_{i-1}^M(m)$,
   - $\pi_{i-1} \circ d_i^{\text{cone}}(m,n) = \pi_{i-1}(-d_{i-1}^M(m), d_i^N(n) - f_{i-1}(m)) = d_{i-1}^M(m)$.

   So $\pi$ is also a morphism of complexes.

3. $\pi(m,n) = 0 \iff m = 0$, so $Z_i(\operatorname{cone} f_\bullet) = 0 \oplus N = \operatorname{im}(x_i)$. So the sequence $0 \mapsto N_i \overset{x_i}{\longmapsto} (\operatorname{cone} f)_i \overset{\pi_i}{\longmapsto} M[-1]_i \mapsto 0$ is exact for all $i$.

In conclusion, the sequence $0 \mapsto N_\bullet \overset{x_\bullet}{\longmapsto} \operatorname{cone} f_\bullet \overset{\pi_\bullet}{\longmapsto} M_\bullet[-1] \mapsto 0$ is exact. □

**Proposition 2.2.** *The short exact sequence* $0 \mapsto N_\bullet \mapsto \mathit{cone}\, f_\bullet \mapsto M_\bullet[-1] \mapsto 0$ *induces a long exact sequence*

$$\cdots \mapsto H_n(M_\bullet) \xmapsto{H_n(f_\bullet)} H_n(N_\bullet) \mapsto H_n(\mathit{cone}\, f_\bullet) \mapsto H_{n-1}(M_\bullet) \xmapsto{H_{n-1}(f_\bullet)} H_{n-1}(N_\bullet) \mapsto \cdots$$

*Proof.* Proposition 2.2, implies that from the sequence $0 \mapsto N_\bullet \mapsto \operatorname{cone} f_\bullet \mapsto M_\bullet[-1] \mapsto 0$ we obtain

$$\cdots \mapsto H_{n+1}(M_\bullet[-1]) \overset{\partial}{\mapsto} H_n(N_\bullet) \mapsto H_n(\operatorname{cone} f_\bullet) \mapsto H_n(M_\bullet[-1]) \overset{\partial}{\mapsto} H_{n-1}(N_\bullet) \mapsto \cdots$$

We know that $H_n(M_\bullet[-1]) = H_{n-1}(M_\bullet)$, so it is enough to show that $\partial_n = H_n(f)$. For that, let $\overline{m} \in H_{n+1}(M[-1]) = H_n(M)$,

$$\begin{aligned}
\partial(\overline{m}) &= \overline{x^{-1} \circ d^{\text{cone}} \circ \pi^{-1}(m)} \\
&= \overline{x^{-1} \circ d^{\text{cone}}(-m, 0)} \\
&= \overline{x^{-1}(0, f(m))} \\
&= \overline{f(m)}
\end{aligned}$$

□

**Corollary 2.3.** *For all complexes of A-module morphisms M and N, a chain complex of A-module morphisms* $\alpha : M \to N$ *is a quasi-isomorphism if and only if the cone of* $\alpha$*, cone(*$\alpha$*), is homologically trivial.*

# Chapter 3

# Resolutions and homological dimensions in Comp $(A$-Mod$)$

After having defined different resolutions for the A-Mod category, we will now generalize these constructions to the category comp A-Mod.

## 3.1 Resolutions

In homological algebra, a resolution is any morphism that preserves homology. This type of morphism is called a quasi-isomorphism or a homology isomorphism. And they are taken in the category of complexes of a given additive category. In our case, it is the category $A$-Mod which is abelian, hence additive.

### 3.1.1 The functors Hom and $\otimes$

#### 3.1.1.1 The functor Hom

**Definition 3.1.** $\mathrm{Hom}_\bullet(M_\bullet, N_\bullet)$ is defined as follows:

- $\mathrm{Hom}_n(M_\bullet, N_\bullet) = \prod_{i\in Z} \hom(M_i, M_{i+n})$
- $d_n\left((f_i : M_i \mapsto N_{i+n})_{i\in Z}\right) = (d^N_{i+n} \circ f_i + (-1)^{n-1} f_{i-1} \circ d^M_i : M_i \mapsto N_{i+n-1})_{i\in Z}$

*Remark.* We have

- $f_\bullet \in Z_0\left(\mathrm{Hom}(M_\bullet, N_\bullet)\right) \implies \forall n \in \mathbb{Z},\ d^N_n \circ f_n - f_{n-1} \circ d^M_n = 0$, so $f_\bullet$ is a complex morphism.
- $f_\bullet \in B_0\left(\mathrm{Hom}(M_\bullet, N_\bullet)\right) \implies \exists (s_n : M_n \mapsto N_{n+1})_{n\in\mathbb{Z}}$ such that $\forall n \in \mathbb{Z}$, $g_n = s_{n-1} \circ d^M_n + d^N_{n+1} \circ s_n$.

So $H_0\left(\mathrm{Hom}(M_\bullet, N_\bullet)\right)$ consists of the homotopy classes of complex morphisms from $M_\bullet$ to $N_\bullet$.

**Proposition 3.1.** *Let the complexes $M_\bullet$, $N_\bullet$ and $P_\bullet$, and the complex morphism $f_\bullet : M_\bullet \mapsto N_\bullet$. Then $Hom(P_\bullet, cone\,(f_\bullet)) = cone\,(Hom(P_\bullet, f_\bullet))$.*

*Proof.*

$$(\operatorname{cone}\operatorname{Hom}(P_\bullet, f_\bullet))_n = \operatorname{Hom}(P_\bullet, M_\bullet)_{n-1} \oplus \operatorname{Hom}(P_\bullet, N_\bullet)_n$$

- For $d_n^{\text{cone}} : (\operatorname{cone}\operatorname{Hom}(P_\bullet, f_\bullet))_n \mapsto (\operatorname{cone}\operatorname{Hom}(P_\bullet, f_\bullet))_{n-1}$. Let $((\alpha_i)_{i\in Z}, (\beta_i)_{i\in Z}) \in (\operatorname{cone}\operatorname{Hom}(P_\bullet, f_\bullet))_n$, then for all $i \in Z$, $\alpha_i : P_i \mapsto M_{i+n-1}$, $\beta_i : P_i \mapsto N_{i+n}$. And we have:

$$\begin{aligned} d_n^{\text{cone}}\ ((\alpha_i)_{i\in Z}, (\beta_i)_{i\in Z}) =& (-d_{n-1}(\alpha_i)_i, d_n(\beta_i)_i - \operatorname{Hom}(P_\bullet, f)_{n-1}(\alpha_i)_i) \\ =& (-(d^M_{i+n-1} \circ \alpha_i + (-1)^n f_{i-1} \circ d^P_i)_i, \\ & (d^N_{i+n} \circ \beta_i + (-1)^{n-1}\beta_{i-1} \circ d^P_i) - (f_{i+n-1} \circ \alpha_i)_i) \end{aligned}$$

- For $d_n^{\text{Hom}} : \operatorname{Hom}(P_\bullet, \operatorname{cone} f_\bullet)_n \mapsto \operatorname{Hom}(P_\bullet, \operatorname{cone} f_\bullet)_{n-1}$

$$\begin{aligned} \text{Soit } (\gamma_i)_{i\in Z} \in \operatorname{Hom}(P_\bullet, \operatorname{cone} f_\bullet)_n &= \prod_{i\in Z} \hom(P_i, \operatorname{cone} f_{i+n}) \\ &= \prod_{i\in Z} \hom(P_i, M_{i+n-1} \oplus M_{i+n}) \end{aligned}$$

  Then

$$d_n^{\text{Hom}}\left((\gamma_i : P_i \mapsto M_{i+n-1} \oplus M_{i+n})_i\right) = \left(d^{\text{cone}}_{i+n} \circ \gamma_i + (-1)^{n-1}\gamma_{i-1} \circ d^P_i\right)$$

  If we represent $\gamma$ by $\gamma = (\alpha, \beta)$ then

$$\gamma_{i-1} \circ d^P_i = (\alpha_{i-1} \circ d^P_i, \beta_{i-1} \circ d^P_i)$$

$$d^{\text{cone}}_{i+n} \circ \alpha_i = d^{\text{cone}}_{i+n}\,(\alpha_i, \beta_i) = (-d^M_{i+n-1} \circ \alpha_i + (-1)^{n-1}\alpha_{i-1} \circ d^P_i, d^N_{i+n} \circ \beta_i + (-1)^{n-1}\beta_{i-1} \circ d^P_i - f_{i+n-1} \circ \alpha_i)$$

□

#### 3.1.1.2 The functor $\otimes$

**Definition 3.2.** Let $M_\bullet$ and $N_\bullet$ be two complexes of right and left $A$-modules respectively, then their tensor product $M_\bullet \otimes N_\bullet$ is the following complex:

- $(M_\bullet \otimes N_\bullet)_n = \bigoplus_{i+j=n} M_j \otimes_A N_j$
- $\forall m_i \in M_i,\ n_j \in N_j,\ d_n(m_i \otimes n_j) = d^M_i(m_i) \otimes n_j + (-1)^i m_i \otimes d^N_j(n_j)$

*Remark.* Let $M \in A$-Mod, $N \in$ Ab, then $\hom(M, N)$ has a right $A$-module structure by $\forall f \in \hom(M, N)$ and $a \in A$, $fa(m) = f(am)$.

**Proposition 3.2.** *Let $M_\bullet \in \mathcal{C}(Mod\text{-}A)$, $N_\bullet \in \mathcal{C}\,((A, B)\text{-}Mod)$ and $T_\bullet \in Mod\text{-}\mathcal{B}$ (with $(A, B)$-Mod the category of $(A, B)$ bimodules). Then:*

$$Hom_B(M_\bullet \otimes_A N_\bullet, T_\bullet) \simeq Hom_A\,(M_\bullet, Hom_B(N_\bullet, T_\bullet))$$

*Proof.* $\forall n$

$$\begin{aligned}
\mathrm{Hom}_B(M_\bullet \otimes_A N_\bullet, T_\bullet)_n &= \prod_i \mathrm{hom}_B\left((M_\bullet \otimes_A N_\bullet)_i, T_{i+n}\right) \\
&= \prod_i \mathrm{hom}_B\left(\bigoplus_j M_j \otimes_A N_{i-j}, T_{i+n}\right) \\
&= \prod_i \prod_j \mathrm{hom}_B(M_j \otimes_A N_{i-j}, T_{i+n}) \\
&= \prod_i \prod_j \mathrm{hom}_A\left(M_j, \mathrm{hom}_B(N_{i-j}, T_{i+n})\right) \\
&= \prod_j \mathrm{hom}_A\left(M_j, \prod_i \mathrm{hom}_B(N_{i-j}, T_{i+n})\right) \\
&= \prod_j \mathrm{hom}_A\left(M_j, \mathrm{Hom}_B(N_\bullet, T_\bullet)_{j+n}\right) \\
&= \mathrm{Hom}_A\left(M_\bullet, \mathrm{Hom}_B(N_\bullet, T_\bullet)\right)_n
\end{aligned}$$

□

### 3.1.2 Projective and injective resolutions

#### 3.1.2.1 K-projective complex

**Definition 3.1.** The complex $P_\bullet$ is K-projective when for any acyclic complex $M_\bullet$, $\mathrm{Hom}(P_\bullet, M_\bullet)$ is acyclic.

**Proposition 3.1.** *Let $P_\bullet$ be a complex, such that for each $i \in \mathbb{Z}$, $P_i$ is projective, if $P_\bullet$ is bounded below then $P_\bullet$ is K-projective.*

*Proof.* Let the complex

$$\cdots \mapsto P_{b+2} \mapsto P_{b+1} \mapsto P_b \mapsto 0$$

such that for all $i$, $P_i$ is projective. Let $M_\bullet$ be an acyclic complex;

$$\mathrm{Hom}(P_\bullet, M_\bullet)_n = \prod_{i\in\mathbb{Z}} \mathrm{hom}(P_i, M_{i+n})$$

and

$$d_n^{\mathrm{hom}}((\alpha_i : P_i \mapsto M_{i+n})_{i\in\mathbb{Z}}) = (d_{i+n}^M \circ \alpha_i + (-1)^{n-1}\alpha_{i-1} \circ d_i^P)_{i\in\mathbb{Z}}$$

We know that $M_{i+n} = M[n]$ and $d_{i+n}^M = (-1)^n d_i^{M[n]}$, so we can write

$$d_n^{\mathrm{Hom}}\left((\alpha_i)_{i\in\mathbb{Z}}\right) = \left((-1)^n (d_i^M \circ \alpha_i - \alpha_{i-1} \circ d_i^P)\right)$$

Suppose that $d_n^{\mathrm{Hom}}((\alpha_i)_{i\in\mathbb{Z}}) = 0 \implies \forall i \in \mathbb{Z},\ d_i^M \circ \alpha_i - \alpha_{i-1} \circ d_i^P = 0$. So $\alpha_\bullet : P_\bullet \mapsto M_\bullet$ is a morphism of complexes. It is therefore sufficient to show that $\alpha_\bullet$ is null-homotopic.

$$\begin{tikzcd}
\cdots \arrow[r, "d^P_{b+3}"] & P_{b+2} \arrow[r, "d^P_{b+2}"] \arrow[d, "\alpha_{b+2}"] & P_{b+1} \arrow[r, "d^P_{b+1}"] \arrow[d, "\alpha_{b+1}"] & P_b \arrow[r] \arrow[d, "\alpha_b"] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r, "d^M_{b+3}"] & M_{b+2} \arrow[r, "d^M_{b+2}"] & M_{b+1} \arrow[r, "d^M_{b+1}"] & M_b \arrow[r, "d^M_b"] & M_{b-1} \arrow[r, "d_{b-1}b^M"] & \cdots
\end{tikzcd}$$

$d_b^M \circ \alpha_b = 0 \implies \operatorname{im}\alpha_b \subseteq \ker d_b^M = Z_b(M)$. So $\operatorname{im}\alpha_b \subseteq B_b(M)$ gives the following diagram:

$$\begin{tikzcd}
 & P_b \arrow[d, "\alpha_b"] \arrow[dl, dashed] \\
M_{b+1} \arrow[r, two heads, "d^M_{b+1}"] & B_b(M)
\end{tikzcd}$$

and $P_b$ projective $\implies \exists s_b$ such that $d_{b+1}^M \circ s_b = \alpha_b$.

Then

$$\begin{aligned}
d_{b+1}^M \circ (\alpha_{b+1} - s_b \circ d_{b+1}^P) &= d_{b+1}^M \circ \alpha_{b+1} - d_{b+1}^M \circ s_b \circ d_{b+1}^P \\
&= \alpha_b \circ d_{b+1}^P - \alpha_b \circ d_{b+1}^P = 0
\end{aligned}$$

. Consequently $\operatorname{im}(\alpha_{b+1} - s_b \circ d_{b+1}^P) \subseteq B_{b+1}(M)$, and so $\exists s_{b+1} : P_{b+1} \mapsto M_{b+2}$ such that $d_{b+2}^M \circ s_{b+1} = \alpha_{b+1} - s_b \circ d_{b+1}^P$.

In the same way:

$$\begin{aligned}
d_{b+2}^M \circ (\alpha_{b+2} - s_{b+1} \circ d_{b+2}^P) &= d_{b+2}^M \circ \alpha_{b+2} - d_{b+2}^M \circ s_{b+1} \circ d_{b+2}^P \\
&= \alpha_{b+1} \circ d_{b+2}^P - \alpha_{b-1} \circ d_{b+2}^P + s_b \circ d_{b+1}^P \circ d_{b+2}^P \\
&= 0
\end{aligned}$$

consequently $\exists s_{b+2}$ such that $\alpha_{b+2} = d_{b+3}^M \circ s_{b+2} + s_{b+1} \circ d_{b+2}^P$.

There exists therefore $(s_i)_{i\in\mathbb{Z}}$ with $s_i = 0$, $\forall i < b$, then $\forall i$, $\alpha_i = d_{i+1}^M \circ s_i + s_{i-1} \circ d_i^P$. $(\alpha_\bullet)_{i\in\mathbb{Z}}$ is null-homotopic. Indeed, if we do the translation again we have:

$$\alpha_i = (-1)^n d_{i+n+1}^M \circ s_i + s_{i-1} \circ d_i^P = d_{n+1}^{\mathrm{hom}}(((-1)^n s_i)_{i\in\mathbb{Z}})$$

with $s_i : P_i \mapsto M_{i+n+1}$. And so $\hom(P_\bullet, M_\bullet)$ is exact, that is to say $P_\bullet$ is K-projective. □

**Proposition 3.2.** *Let $P_\bullet$ be K-projective, then for any quasi-isomorphism $f_\bullet : M_\bullet \mapsto N_\bullet$, the morphism of complexes $Hom(P_\bullet, f_\bullet) : Hom(P_\bullet, M_\bullet) \mapsto Hom(P_\bullet, N_\bullet)$ is a quasi-isomorphism.*

*Proof.* Since $f_\bullet : M_\bullet \mapsto N_\bullet$ is a quasi-isomorphism, then $\operatorname{cone} f_\bullet$ is exact.

From the morphism $\mathrm{Hom}(P_\bullet, f_\bullet) : \mathrm{Hom}(P_\bullet, M_\bullet) \mapsto \mathrm{Hom}(P_\bullet, N_\bullet)$ we obtain the short exact sequence of complexes:

$$0 \mapsto \mathrm{Hom}(P_\bullet, N_\bullet) \mapsto \operatorname{cone}\mathrm{Hom}(P_\bullet, f_\bullet) \mapsto \mathrm{Hom}(P_\bullet, M_\bullet)[-1] \mapsto 0$$

which induces the long exact sequence of homology

$$\begin{tikzcd}
\cdots \arrow[r] & H_{n+1}(\operatorname{cone}\operatorname{Hom}(P_\bullet, f_\bullet)) \arrow[d] & & \\
& H_n(\operatorname{Hom}(P_\bullet, M_\bullet)) \arrow[r, "H_n(\operatorname{Hom}(P_\bullet\ f_\bullet))"] \arrow[d, equal] & H_n(\operatorname{Hom}(P_\bullet, N_\bullet)) \arrow[d] & \\
& H_{n+1}(\operatorname{Hom}(P_\bullet, M_\bullet)[-1]) & H_{n+1}(\operatorname{cone}\operatorname{Hom}(P_\bullet, f_\bullet)) \arrow[r] & \cdots
\end{tikzcd}$$

With, $H_n(\operatorname{cone}\operatorname{Hom}(P_\bullet, f_\bullet)) = H_n(\operatorname{Hom}(P_\bullet, \operatorname{cone} f))$. Since cone $f$ is exact then $H_n(\operatorname{Hom}(P_\bullet, \operatorname{cone} f_\bullet)) = 0$. So

$$H_n(\operatorname{Hom}(P_\bullet, f_\bullet)) : H_n(\operatorname{Hom}(P_\bullet, M_\bullet)) \mapsto H_n(\operatorname{Hom}(P_\bullet, N_\bullet))$$

is an isomorphism. □

#### 3.1.2.2 K-projective resolution

**Definition 3.2.** Let $M_\bullet$ be a complex of $A$-modules, a K-projective resolution of $M_\bullet$ is the data of a K-projective complex $P_\bullet$ and a quasi-isomorphism $P_\bullet \mapsto M_\bullet$.

**Proposition 3.3.** *Any complex $M_\bullet$ bounded below admits a K-projective resolution.*

*Proof.* There exists a surjection $\alpha_b : P_b \mapsto M_b$ such that $P_b$ is projective. Then let the module

$$L_{b+1} = \left\{(p, m) \in P_b \times M_{b+1} \mid \alpha_b(p) = d^M_{b+1}(m)\right\}$$

and the two canonical morphisms $\omega : L_{b+1} \mapsto M_{b+1}$ and $x : L_{b+1} \mapsto P_b$. For any $m \in M_{b+1}$, since $\alpha_b$ is surjective, there exists $p \in P_b$ such that $\alpha_b(p) = d^M_{b+1}(m)$. So $\omega$ is surjective. And we have $\alpha_b \circ x = d^M_{b+1} \circ w$. Then let the surjection with a projective module $P_{b+1}$, $P_{b+1} \mapsto L_{b+1}$, and the morphisms $\alpha_{b+1} : P_{b+1} \mapsto L_{b+1} \overset{w}{\mapsto} M_{b+1}$, and $d^M_{b+1} : P_{b+1} \mapsto L_{b+1} \overset{x}{\mapsto} P_b$.

$$\begin{tikzcd}
P_{b+1} \arrow[rr, "d^P_{b+1}"] \arrow[dr, two heads] \arrow[dd, two heads, "\alpha_{b+1}"] & & P_b \arrow[r] \arrow[dd, two heads, "\alpha_b"] & 0 \\
& L_{b+1} \arrow[ur, "x"] \arrow[dl, two heads, "\omega"] & & \\
M_{b+1} \arrow[rr, "d^M_{b+1}"] & & M_b \arrow[r] & 0
\end{tikzcd}$$

Let

$$\begin{aligned}
\overline{\alpha_b} : P_b &\mapsto M_b/B_b(M) = H_b(M) \\
p &\mapsto \overline{\alpha_b(p)}
\end{aligned}$$

Then

$$\begin{aligned}\overline{\alpha_b(p)} = 0 &\iff \alpha_b(p) \in B_b(M)\\ &\iff \exists m \in M_{b+1} \mid \alpha_b(p) = d^M_{b+1}(m)\\ &\iff p \in \operatorname{im} x = \operatorname{im} d^P_{b+1}\end{aligned}$$

So $\ker \overline{\alpha_b} = B_b(P)$, in other words $H_b(P) \simeq H_b(M)$. Then we also have

$$\begin{aligned}m \in Z_{b+1}(M) &\implies d^M_{b+1}(m) = 0\\ &\implies d^M_{b+1}(m) = \alpha_b(0)\\ &\implies (0, m) \in L\\ &\implies \exists p \in P_{b+1} \mid \alpha_{b+1}(p) = m \wedge d^P_{b+1}(p) = 0\\ &\implies \exists p \in Z_{b+1}(P) \mid \alpha_{b+1}(p) = m\end{aligned}$$

And let $p \in Z_{b+1}(P)$, then $d^M_{b+1}(\alpha_{b+1}(p)) = \alpha_b(d^M_{b+1}(p)) = 0$. So $\forall p \in Z_{b+1}(P), \alpha_{p+1}(p) \in Z_{b+1}(M)$. So $\alpha_{b+1}$ induces a surjective morphism from $Z_{b+1}(P)$ to $Z_{b+1}(M)$.

Then let

$$L_{b+2} = \left\{(p, m) \in Z_{b+1}(P) \times M_{b+2} \mid \alpha_{b+1}(p) = d^M_{b+2}(m) \in Z_{b+1}(M)\right\}$$

and let $P_{b+2} \mapsto L_{b+2}$, $L_{b+2} \overset{\omega}{\mapsto} M_{b+2}$ and $L_{b+2} \overset{x}{\mapsto} P_{b+1}$. In the same way, we construct $\alpha_{b+2}$ surjective and $d^P_{b+2}$ which make the diagram commutative, and induce a surjective morphism from $Z_{b+2}(P)$ to $Z_{b+2}(M)$, and an isomorphism $H_{b+1}(P) \simeq H_{b+1}(M)$.

Thus we obtain a complex $P_\bullet$ of projective objects such that $P_\bullet$ is quasi-isomorphic to $M_\bullet$. So $P_\bullet$ is a K-projective resolution of $M_\bullet$. □

**Proposition 3.4.** *Let the quasi-isomorphism $f_\bullet : N_\bullet \mapsto M_\bullet$, and a projective resolution $r_\bullet : P_\bullet \mapsto M_\bullet$ of $M_\bullet$. Then there exists a morphism of complexes $\alpha_\bullet : P_\bullet \mapsto N_\bullet$ such that $f_\bullet \circ \alpha_\bullet \sim r_\bullet$.*

*Proof.* Since $P$ is K-projective, $\operatorname{Hom}(P_\bullet, f_\bullet) : \operatorname{Hom}(P_\bullet, N_\bullet) \mapsto \operatorname{Hom}(P_\bullet, M_\bullet)$ is a quasi-isomorphism. So $H_0(\operatorname{Hom}(P_\bullet, N_\bullet)) \simeq H_0(\operatorname{Hom}(P_\bullet, M_\bullet))$. Consequently there exists $\alpha_\bullet : P_\bullet \mapsto N_\bullet$ a morphism of complexes such that $f_\bullet \circ \alpha_\bullet \sim r_\bullet$. □

**Corollary 3.5.** *Let the quasi-isomorphism $f_\bullet : N_\bullet \mapsto M_\bullet$, and a projective resolution $r_\bullet : P_\bullet \mapsto M_\bullet$ of $M_\bullet$. Then there exists a quasi-isomorphism $\alpha_\bullet : P_\bullet \mapsto N_\bullet$.*

*Proof.* $f_\bullet \circ \alpha_\bullet \sim r_\bullet$. So since $r_\bullet$ is a quasi-isomorphism, $f_\bullet \circ \alpha_\bullet$ is also one. And since $H(f_\bullet)$ and $H(f_\bullet \circ \alpha_\bullet) = H(f_\bullet) \circ H(\alpha_\bullet)$ are isomorphisms, then $H(\alpha_\bullet)$ is also one. □

**Proposition 3.6.** *Let $P_\bullet \overset{\sim}{\mapsto} M_\bullet$ and $P'_\bullet \overset{\sim}{\mapsto} M_\bullet$ be two K-projective resolutions of $M_\bullet$. Then $P_\bullet$ and $P'_\bullet$ are homotopic.*

*Proof.* Let the quasi-isomorphisms $r_\bullet : P_\bullet \mapsto M_\bullet$ and $r'_\bullet : P'_\bullet \mapsto M_\bullet$. Then there exists $f_\bullet : P_\bullet \mapsto P'_\bullet$ such that $r'_\bullet \circ f_\bullet \sim r_\bullet$.

$$\begin{tikzcd} & P_\bullet \arrow[dl, dashed, "f_\bullet"'] \arrow[d, "r_\bullet"] \\ P'_\bullet \arrow[r, "r'_\bullet"'] & M_\bullet \end{tikzcd}$$

Since $r_\bullet$ and $r'_\bullet$ are quasi-isomorphisms, consequently $f_\bullet$ is a quasi-isomorphism. And there exists $g_\bullet : P'_\bullet \mapsto P_\bullet$ such that $f_\bullet \circ g_\bullet \sim 1_{P'_\bullet}$.

$$\begin{tikzcd} & P'_\bullet \arrow[dl, dashed, "g_\bullet"'] \arrow[d, "1_{P'_\bullet}"] \\ P_\bullet \arrow[r, "f_\bullet"'] & P'_\bullet \end{tikzcd}$$

So $g_\bullet$ is a quasi-isomorphism (because $f_\bullet$ and $1_{P_\bullet}$ are), and there exists $f'_\bullet : P_\bullet \mapsto P'_\bullet$ such that $g_\bullet \circ f'_\bullet \sim 1_{P_\bullet}$.

$$\begin{tikzcd} & P_\bullet \arrow[dl, dashed, "f'_\bullet"'] \arrow[d, "1_{P_\bullet}"] \\ P'_\bullet \arrow[r, "g_\bullet"'] & P_\bullet \end{tikzcd}$$

Thus, $g_\bullet \circ f'_\bullet \sim 1_{P_\bullet}$ and $f_\bullet \circ g_\bullet \sim 1_{P'_\bullet} \implies f_\bullet \sim f_\bullet \circ g_\bullet \circ f'_\bullet \sim f'_\bullet$. Consequently $g_\bullet \circ f_\bullet \sim g_\bullet \circ f'_\bullet \sim 1_{P_\bullet}$. Thus $g_\bullet \circ f_\bullet \sim 1_{P_\bullet}$ and $f_\bullet \circ g_\bullet \sim 1_{P'_\bullet}$ consequently $P_\bullet$ and $P'_\bullet$ are homotopic. $\square$

### 3.1.3 Resolutions of unbounded complexes

**Definition 3.1.** Let $\mathcal{S}$ be a class of complexes of $A$-modules, then:

- A direct (resp. projective) system $(M_i)_{i\in I}$ is said to be $\mathcal{S}$-special if:
  - $I$ is well-ordered,
  - if $i \in I$ has no predecessor then $M_i = \varinjlim_{j<i} M_j$ (resp. $M_i = \varprojlim_{j<i} M_j$)
  - if $i \in I$ has a predecessor $i-1$, then $M_{i-1} \mapsto M_i$ is injective, $C_i = \operatorname{coker}(M_{i-1} \mapsto M_i) \in \mathcal{S}$ and $0 \mapsto M_{i-1} \mapsto M_i \mapsto C_i \mapsto 0$ is split (resp. $M_i \mapsto M_{i-1}$ is surjective, $K_i = \ker(M_i \mapsto M_{i-1}) \in \mathcal{S}$ and $0 \mapsto K_i \mapsto M_i \mapsto M_{i-1} \mapsto 0$ is split).
- $\mathcal{S}$ will be said to be closed for special direct (resp. projective) limits if any $\mathcal{S}$-special direct (resp. inverse) system has a limit in $\mathcal{S}$, and if any complex isomorphic to a complex in $\mathcal{S}$ is in $\mathcal{S}$.

**Lemma 3.1.** *The class of acyclic complexes is closed for special projective limits.*

**Proposition 3.2.** *The class of K-injective complexes is closed for special projective limits.*

*Proof.* Let $(M_i)_I$ be a projective system of complexes of $A$-modules such that:

- $I$ is well-ordered,
- if $i \in I$ has no predecessor then $M_i = \varprojlim_{j<i} M_j$
- if $i \in I$ has a predecessor $i-1$ then $M_i \mapsto M_{i-1}$ is surjective, $K_i = \ker(M_i \mapsto M_{i-1})$ is $K$-injective and $0 \mapsto K_i \mapsto M_i \mapsto M_{i-1} \mapsto 0$ is split.

$\forall T$ an exact complex of $A$-modules, we then have the split short exact sequences $0 \mapsto \mathrm{Hom}(T, K_i) \mapsto \mathrm{Hom}(T, M_i) \mapsto \mathrm{Hom}(T, M_{i-1}) \mapsto 0$ and the projective system $(\mathrm{Hom}(T, M_i))_{i\in I}$ which satisfies $(\star)$. So $\varprojlim_I \mathrm{Hom}(T, M_i) = \mathrm{Hom}(T, \varprojlim_I M_i)$ is exact. so $\varprojlim_I M_i$ is K-injective. □

**Proposition 3.3.** *the class of K-projective complexes is closed for special inductive limits.*

*Proof.* Let $(M_i)_I$ be a direct system of complexes of $A$-modules such that:

- $I$ is well-ordered,
- if $i \in I$ has no predecessor then $M_i = \varinjlim_{j<i} M_j$
- if $i \in I$ has a predecessor $i-1$ then $M_{i-1} \mapsto M_i$ is injective, $C_i = \mathrm{coker}\,(M_{i-1} \mapsto M_i)$ is K-projective and $0 \mapsto M_{i-1} \mapsto M_i \mapsto C_i \mapsto 0$ is split.

$\forall T$ an exact complex of $A$-modules, we then have the split short exact sequences $0 \mapsto \mathrm{Hom}(C_i, T) \mapsto \mathrm{Hom}(M_i, T) \mapsto \mathrm{Hom}(M_{i-1}, T) \mapsto 0$ and the projective system $(\mathrm{Hom}(M_i, T))_{i\in I}$ which satisfies $(\star)$. So $\varprojlim_I \mathrm{Hom}(M_i, T) = \mathrm{Hom}(\varprojlim_I M_i, T)$ is exact. so $\varprojlim_I M_i$ is K-projective. □

**Lemma 3.4.** $\forall M_\bullet \in \mathcal{C}(A\text{-}Mod)$,

- *there exists a direct system of K-projective complexes* $(P^n_\bullet)_{n\geq -1}$ *and a direct system of quasi-isomorphisms* $(f^n_\bullet : P^n_\bullet \mapsto \tau_{\geq -n} M_\bullet)_{n\geq -1}$.
- *There exists a projective system of K-injective complexes* $(I^n_\bullet)_{n\geq 1}$ *and a projective system of quasi-isomorphisms* $(g^n_\bullet : \tau_{\leq n} M_\bullet \mapsto I^n_\bullet)_{n\geq 1}$

*Proof.* The proof is by induction

- For $n = -1$, we set $P^{-1}_\bullet = 0$, and $f^{-1}_\bullet = 0$.
- For $n = 0$, we know that there exists a K-projective complex $P^0_\bullet$ and the resolution $f^0_\bullet : P^0_\bullet \mapsto \tau_{\geq 0} M$.

- Let $n \geq 1$, suppose that there exist the K-projective resolutions $f_\bullet^{-1} : P_\bullet^{-1} \mapsto \tau_{\geq 1} M_\bullet, \cdots, f_\bullet^{n-1} : P_\bullet^{n-1} \mapsto \tau_{\geq -n+1} M_\bullet$. Let $P_\bullet = P_\bullet^{n-1}$ and $N = \tau_{\geq -n} M$.

$$\begin{tikzcd}
\cdots P_{-n+1} \arrow[r] \arrow[d, "f_{-n+1}"'] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots M_{-n+1} \arrow[r, "d^M_{-n+1}"] & \ker d^M_{-n} \arrow[r] & 0 \cdots
\end{tikzcd}$$

and let $f_\bullet : P_\bullet \mapsto N_\bullet$ be the induced complex morphism (it is defined because $d^M_{-n+1} \circ (f_{-n+1} = f^{n-1}_{-n+1} : P_{-n+1} \mapsto \ker d^M_{-n+1}) = 0$).

$$\operatorname{cone}(f_\bullet) = \cdots \mapsto P_{-n+1} \oplus M_{-n+2} \mapsto M_{-n+1} \mapsto M_n \mapsto 0 \cdots$$

there exists a K-projective resolution of $\operatorname{cone}(f)[1]$, $g : Q_\bullet \mapsto \operatorname{cone}(f)[1]$. We write $g = (g'_\bullet : Q_\bullet \mapsto P_\bullet, g''_\bullet : Q_\bullet \mapsto M_\bullet[1])$.

$$\begin{tikzcd}
\cdots \arrow[r] & Q_{-n+1} \arrow[r] \arrow[d, "g_{-n+1}"] & Q_{-n} \arrow[r] \arrow[d, "g_{-n}"] & Q_{-n-1} \arrow[r] \arrow[d, "g_{-n-1}"] & 0 \cdots \\
\cdots \arrow[r] & P_{-n+1} \oplus M_{-n+2} \arrow[r] & M_{-n+1} \arrow[r] & M_{-n} \arrow[r] & 0 \cdots
\end{tikzcd}$$

$$\begin{tikzcd}
\cdots \arrow[r] & Q_{-n+1} \arrow[r] \arrow[d, "g'"] & Q_{-n} \arrow[r] \arrow[d] & Q_{-n-1} \arrow[r] \arrow[d] & 0 \cdots \\
\cdots \arrow[r] & P_{-n+1} \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \cdots
\end{tikzcd}$$

$\forall n$, $g_{n-1} \circ d^Q_n = (g'_{n-1} \circ d^Q_n, g''_{n-1} \circ d^Q_n)$ and $d_n^{\operatorname{cone}(f)[1]} \circ g_n(q) = -d^{\operatorname{cone}(f)}_{n+1}(g'_n(q), g''_n(q)) = \left(d^P_n \circ g'_n(q), f_n \circ g'_n(q) - d^M_{n+1} \circ g''_n(q)\right)$. And $g_{n-1} \circ d^Q_n(q) = d_n^{\operatorname{cone}(f)[1]} \circ g_n(q)$

$$\implies g'_{n-1} \circ d^Q_n = d^P_n \circ g'_n \text{ and } g''_{n-1} \circ d^Q_n = f_n \circ g'_n - d^M_{n+1} \circ g''_n$$

So $g'_\bullet$ is a morphism of complexes. and let

$$\begin{aligned} h_\bullet : \operatorname{cone}(-g'_\bullet) &\longmapsto M_\bullet \\ (x, y) &\longmapsto g''(x) + f(y) \end{aligned}$$

$$\begin{tikzcd}
\cdots \arrow[r] & Q_{-n} \oplus P_{-n+1} \arrow[r] \arrow[d] & Q_{-n-1} \arrow[r] \arrow[d] & 0 \cdots \\
\cdots \arrow[r] & M_{-n+1} \arrow[r] & M_{-n} \arrow[r] & 0 \cdots
\end{tikzcd}$$

Then

$$\begin{aligned}
h_n \circ d^{\operatorname{cone}}_{n+1}(x, y) &= h_n\left(-d^Q_n(x), g'_n(x) + d^P_{n+1}(y)\right) \\
&= -g''_{n-1} \circ d^Q_n(x) + f_n\left(g'_n(x) + d^P_{n+1}(y)\right) \\
d^M_{n+1} \circ h_{n+1}(x, y) &= d^M_{n+1}\left(g''_n(x) + f_{n+1}(y)\right) \\
&= d^M_{n+1} \circ g''_n(x) + d^M_{n+1} \circ f_{n+1}(y) \\
&= -g''_{n-1} \circ d^Q_n(x) + f_n \circ g'_n(x) + f_n \circ d^P_{n+1}(y)
\end{aligned}$$

$h_\bullet$ is a morphism of complexes. And $(\operatorname{cone} h)_n = (Q_{n-2} \oplus P_{n-1}) \oplus M_n$ is equipped with the derivative:

$$\begin{aligned} d_n^{\operatorname{cone} h}(x,y,z) &= \Big(-d_{n-1}^{\operatorname{cone} -g'}(x,y), -h_{n-1}(x,y) + d_n^M(z)\Big) \\ &= \Big(d_{n-2}^Q(x), -g'_{n-2}(x) - d_{n-1}^P(y), \\ &\qquad - g''_{n-2}(x) - f_{n-1}(y) + d_n^M(z)\Big) \end{aligned}$$

And for $\operatorname{cone} g$, we have $(\operatorname{cone} g)_n = Q_{n-1} \oplus (P_n \oplus M_{n+1})$ and the derivative gives:

$$\begin{aligned} d_n^{\operatorname{cone} g}(x,y,z) &= \Big(-d_{n-1}^Q(x), -g_{n-1}(x) + d_n^{\operatorname{cone} f[1]}(y,z)\Big) \\ &= \Big(-d_{n-1}^Q(x), \big(-g'_{n-1}(x), -g''_{n-1}(x)\big) - d_{n+1}^{\operatorname{cone} f}(y,z)\Big) \\ &= \Big(-d_{n-1}^Q(x), \big(-g'_{n-1}(x), -g''_{n-1}(x)\big) \\ &\qquad + \big(d_n^P(y), f_n(y) - d_{n+1}^M(z)\big)\Big) \\ &= \Big(-d_{n-1}^Q(x), -g'_{n-1}(x) + d_n^P(y), \\ &\qquad - g''_{n-1}(x) + f_n(y) - d_{n+1}^M(z)\Big) \end{aligned}$$

So $\operatorname{cone} h[1] = \operatorname{cone} -g$, and since $g$ is a quasi-isomorphism, then $h$ is also one.

□

**Theorem 3.5.** *Any complex of A-modules admits a K-projective resolution, and a K-injective resolution.*

*Proof.*

- For the K-projective resolution it is sufficient to apply the preceding lemma. Given that the inductive limit preserves quasi-isomorphisms, $\varinjlim_n P_\bullet^n \longmapsto \varinjlim_n \tau_{\geq -n} M_\bullet = M_\bullet$ is a quasi-isomorphism.
- So there exists, $(I_\bullet^n)_{n\geq -1}$ a projective system of K-injective complexes and the projective system of quasi-isomorphisms $(g^n : I_\bullet^n \mapsto \tau_{\leq n} M_\bullet)_{n\geq 1}$. $(I_\bullet^n)_{n\geq -1}$ satisfies $(\star)$ so $\forall n \geq 0$;

$$\begin{tikzcd} H_n(M_\bullet) \arrow[r, "f"] \arrow[d, "H_n(\pi_{n+1}^M)"'] & H_n(I_\bullet) \arrow[d, "H_n(\pi_{n+1}^I)"] \\ H_n(\tau_{\leq n+1} M_\bullet) \arrow[r, "H_n(g_{n+1})"] & H_n(I_\bullet^{n+1}) \end{tikzcd}$$

so $H_n(f)$ is an isomorphism, as the three remaining morphisms are isomorphisms.

And $\forall n < 0$, $H_n(M_\bullet) \simeq H_n(I_0) \simeq H_n(M_\bullet)$.

□

### 3.1.4 Flat resolution

**Definition 3.1.** We say that the complex $P_\bullet$ is K-flat, if $\forall M_\bullet$ acyclic, $P_\bullet \otimes_A M_\bullet$ is acyclic.

**Proposition 3.1.** *Let $P_\bullet \in \mathcal{C}(Mod\text{-}A)$, if $P_\bullet$ is bounded below, and that $\forall i$, $P_i$ is flat, then $P_\bullet$ is K-flat.*

**Proposition 3.2.** *Any K-projective complex is K-flat.*

*Proof.* Suppose that $P_\bullet$ is K-projective, let $I_\bullet$ be a K-injective complex and $M_\bullet$ an acyclic complex. Then $\mathrm{Hom}(P_\bullet \otimes M_\bullet, I_\bullet) \simeq \mathrm{Hom}(P_\bullet, \mathrm{Hom}(M_\bullet, I_\bullet))$ is acyclic, so $P_\bullet \otimes M_\bullet$ is acyclic. □

**Corollary 3.3.** *Any complex admits a K-flat resolution.*

## 3.2 Homological dimensions

We introduce the following ext and tor functors, to facilitate the calculations in this section.

### 3.2.1 The functors Ext and Tor

#### 3.2.1.1 The functor Ext

Let the complex $M_\bullet$, $P_\bullet \mapsto M_\bullet$ and $P'_\bullet \mapsto M_\bullet$ be two K-projective resolutions of $M_\bullet$. Then we know that $P_\bullet$ and $P'_\bullet$ are homotopic.

**Proposition 3.1.** *Let $P_\bullet$ and $P'_\bullet$ be two homotopic K-projective complexes, then for any complex $N_\bullet$, $Hom(P_\bullet, N_\bullet)$ is quasi-isomorphic to $Hom(P'_\bullet, N_\bullet)$.*

*Proof.* Let the homotopy equivalences $f_\bullet : P_\bullet \mapsto P'_\bullet$ and $f'_\bullet : P'_\bullet \mapsto P_\bullet$. So $f_\bullet \circ f'_\bullet \sim 1_{P'_\bullet}$ and $f'_\bullet \circ f_\bullet \sim 1_{P_\bullet}$. And we have the complex morphism $\mathrm{Hom}(f_\bullet, N_\bullet) : \mathrm{Hom}(P'_\bullet, N_\bullet) \mapsto \mathrm{Hom}(P_\bullet, N_\bullet)$. We want to show that this is a quasi-isomorphism. Without loss of generality we consider the morphism at index 0,

$$\begin{aligned} H_0\left(\mathrm{Hom}(f_\bullet, N_\bullet)\right) : H_0\left(\mathrm{Hom}(P'_\bullet, N_\bullet)\right) &\mapsto H_0\left(\mathrm{Hom}(P_\bullet, N_\bullet)\right) \\ \overline{\alpha_\bullet} &\mapsto \overline{\alpha_\bullet \circ f_\bullet} \end{aligned}$$

- Let $\beta_\bullet : P_\bullet \mapsto N_\bullet$, so $\beta_\bullet \sim \beta_\bullet \circ f'_\bullet \circ f_\bullet$, with $\beta_\bullet \circ f'_\bullet \in \mathrm{hom}(P'_\bullet, N_\bullet)$. So $H_0\left(\mathrm{Hom}(f_\bullet, N_\bullet)\right)$ is surjective.
- Let $\alpha_\bullet$ and $\alpha'_\bullet : P'_\bullet \mapsto N_\bullet$ such that $\alpha_\bullet \circ f_\bullet \sim \alpha'_\bullet \circ f_\bullet$. then $\alpha_\bullet \sim \alpha_\bullet \circ f_\bullet \circ f'_\bullet \sim \alpha'_\bullet \circ f_\bullet \circ f'_\bullet \sim \alpha'_\bullet$. So $H_0\left(\mathrm{Hom}(f_\bullet, N_\bullet)\right)$ is injective.

So $H_0\left(\mathrm{Hom}(f_\bullet, N_\bullet)\right)$ is an isomorphism. We conclude that $\mathrm{Hom}(f_\bullet, N_\bullet)$ is a quasi-isomorphism. □

**Proposition 3.2.** *Let $I_\bullet$ and $I'_\bullet$ be two homotopic K-injective complexes, then for any complex $M_\bullet$, $Hom(M_\bullet, I_\bullet)$ is quasi-isomorphic to $Hom(M_\bullet, I'_\bullet)$.*

**Proposition 3.3.** *Let $M_\bullet$, $N_\bullet$ be two complexes, $P_\bullet \mapsto M_\bullet$ a K-projective resolution of $M_\bullet$ and $N_\bullet \mapsto I_\bullet$ a K-injective resolution of $N_\bullet$. Then we have the quasi-isomorphisms*

$$Hom(P_\bullet, N_\bullet) \simeq Hom(P_\bullet, I_\bullet) \simeq Hom(M_\bullet, I_\bullet)$$

**Definition 3.1.** Given the complexes $M_\bullet$ and $N_\bullet$, we take any K-projective resolution of $M_\bullet$ or any K-injective resolution of $N_\bullet$. We define $\mathrm{Ext}^n_A(M_\bullet, N_\bullet)$ as being $H_{-n}\left(\mathrm{Hom}(P_\bullet, N_\bullet)\right)$ or $H_{-n}\left(\mathrm{Hom}(M_\bullet, I_\bullet)\right)$.

#### 3.2.1.2 The functor Tor

**Proposition 3.4.** *Let $M'_\bullet$, $N'_\bullet$ two complexes with $P$ and $P'$ K-projective resolutions of $M'$ and $F$, $F'$ K-flat resolutions of $N'$, then $\forall i \in \mathbb{Z}$*

$$H_i(P_\bullet \otimes_A F_\bullet) \cong H_i(P'_\bullet \otimes_A F'_\bullet)$$

**Definition 3.2.** Given the complexes $M'_\bullet$ and $N'_\bullet$, we take any K-projective resolution of $M'_\bullet$ or any K-flat resolution of $N'_\bullet$. We define $\mathrm{Tor}^A_n(M'_\bullet, N'_\bullet)$ as being $H_n\left(P_\bullet \otimes_A F_\bullet\right)$.

### 3.2.2 The K-projective, K-injective and K-flat dimensions

All the element developed in this subsection can be found in Homological dimensions of unbounded complexes[1].

**Definition 3.1.** Let $M_\bullet \in \mathcal{C}(A\text{-Mod})$. For $n \in \mathbb{Z}$, if $M_\bullet$ admits a projective resolution $P_\bullet \mapsto M_\bullet$ (resp. injective resolution $M_\bullet \mapsto I_\bullet$) such that $P_i = 0$, $\forall i > n$ (resp. $I_i = 0$, $\forall i < -n$), then we say that the projective dimension (resp. injective dimension) of $M_\bullet$ is at most $n$. The projective dimension of $M_\bullet$ is defined by

$$\mathrm{pd}_A(M_\bullet) = \inf_{P \in \mathbb{P}} \left\{ \sup_{i \in \mathbb{Z}} \{i | P_i \neq 0\} \right\}$$

$\mathbb{P}$ is the set of projective resolutions of $M_\bullet$. The injective dimension is defined by

$$\mathrm{id}_A(M_\bullet) = \inf_{I \in \mathbb{I}} \left\{ \sup_{i \in \mathbb{Z}} \{-i | I_i \neq 0\} \right\}$$

$\mathbb{I}$ is the set of injective resolutions of $M_\bullet$.

**Definition 3.2.** Let $M_\bullet \in \mathcal{C}(A - Mod)$. For $n \in \mathbb{Z}$, if $M_\bullet$ admits a projective resolution $P_\bullet \mapsto M_\bullet$ such that $\forall i > n$, $P_i = 0$ then we say that the flat dimension of $M_\bullet$ is at most $n$. It is defined by

$$\mathrm{fd}_A(M_\bullet) = \inf_{P \in \mathbb{F}} \left\{ \sup_{i \in \mathbb{Z}} \{i | P_i \neq 0\} \right\}$$

$\mathbb{F}$ is the set of flat resolutions of $M_\bullet$.

*Remark.* These different dimensions satisfy the following properties.

1. If $M_\bullet$ and $M'_\bullet$ are quasi-isomorphic then they have the same projective, injective and flat dimensions.
2. $\operatorname{pd}_A(M_\bullet) \geq \sup M_\bullet$, $\operatorname{id}_A(M_\bullet) \geq -\inf M_\bullet$, and $\operatorname{fd}_A(M_\bullet) \geq \sup M_\bullet$.
3. If $M_\bullet$ is acyclic then $\operatorname{pd}_A(M_\bullet) = \operatorname{id}_A(M_\bullet) = \operatorname{fd}_A(M_\bullet) = -\infty$.

**Proposition 3.1.** *For any complex $M_\bullet$ the following conditions are equivalent:*

1. *$\operatorname{pd}_A(M_\bullet) \leq n$.*
2. *For any complex $N_\bullet$, $\forall i > n - \inf N_\bullet$, $Ext^i_A(M_\bullet, N_\bullet) = 0$.*
3. *For any module $N$, $\forall i > n-1$, $Ext^i_A(M_\bullet, N) = 0$.*
4. *$\forall i > n$, $H_i(M) = 0$, and for any K-projective resolution of $M_\bullet$, $coker(d^P_{n+1})$ is projective.*
5. *Any K-projective resolution $P_\bullet$ of $M_\bullet$ is isomorphic to a coproduct of K-projective complexes $P'_\bullet \bigoplus P''_\bullet$ such that $P'_i = 0$, $\forall i > n$ and $P''_\bullet$ is contractible.*
6. *Any K-projective resolution $P_\bullet$ of $M_\bullet$ is isomorphic to a coproduct of K-projective complexes $P'_\bullet \bigoplus P''_\bullet$ such that $P'_i = 0$, $\forall i > n$ and $P''_\bullet$ is contractible.*

*Proof.*

$1 \implies 2$ $\operatorname{pd}_A(M_\bullet) \leq n$ implies that there exists a K-projective resolution $P_\bullet \mapsto M_\bullet$ such that $P_i = 0$, $\forall i > n$. Let a complex $N_\bullet$, let $N' = \tau_{\geq \inf N}$ so $N_\bullet$ is quasi-isomorphic to $N'$ and we have then

$$\operatorname{Ext}^i_A(M_\bullet, N_\bullet) \cong H_i\left(\operatorname{Hom}(P_\bullet, N_\bullet)\right) \cong H_i\left(\operatorname{Hom}(P_\bullet, N'_\bullet)\right)$$

So $\forall i > n - \inf N$, $\operatorname{Hom}_A(P_\bullet, N'_\bullet)_{-i} = 0$,

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] & P_n \arrow[r] \arrow[drrr] & \cdots \arrow[r] & P_{\inf N} \arrow[r] & P_{\inf N - 1} \arrow[r] & \cdots \\
\cdots \arrow[r] & N'_{n+1} \arrow[r] & N'_n \arrow[r] & \cdots \arrow[r] & N'_{\inf N} \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

whence $\operatorname{Ext}^n_i(M_\bullet, N_\bullet) \cong H_{-i}\left(\operatorname{Hom}_A(P_\bullet, N'_\bullet)\right) = 0$.

$2 \implies 3$ Let $N$ be an $A$-module, then $\inf N = -1$, so $\forall i > n+1$, $\operatorname{Ext}^i_A(M_\bullet, N) = 0$.

$3 \implies 4$ $\forall i > n$, $H_i(M) = 0$, Let $P \mapsto M_\bullet$ be a K-projective resolution of $M_\bullet$, then $\forall i > n$, $H_i(P_\bullet) = 0$ and $\cdots \mapsto P_{n+1} \xmapsto{d^P_{n+1}} P_n \mapsto \operatorname{coker} d^P_{n+1} \mapsto 0$ is

a projective resolution (K-projective if we consider them as complexes) of $\operatorname{coker} d^P_{n+1}$. Let the $A$-module $N$, then $\forall i > 0$

$$\begin{aligned}\operatorname{Ext}^1_A\left(\operatorname{coker} d^P_{n+1}, N\right) &\cong H_{-1}\left(\operatorname{Hom}_A(P[n], N)\right)\\ &\cong \operatorname{Hom}_A\left(H_{n+i}(P), N\right)\\ &\cong \operatorname{Hom}_A\left(0, N\right) = 0\end{aligned}$$

So $\operatorname{coker} d^P_{n+1}$ is projective.

$4 \implies 5$ Let $P_\bullet$ be a K-projective resolution of $M$ then we have the exact sequence:

$$0 \mapsto P''_\bullet \mapsto P_\bullet \xmapsto{\tau_{\leq n}} \overline{P}_\bullet \mapsto 0$$

with $\overline{P}_\bullet = \tau_{\leq n}(P_\bullet)$. We also have the exact sequence

$$0 \mapsto B_n(P_\bullet) \mapsto P_n \mapsto \operatorname{coker} d^P_{n+1} \mapsto 0$$

Since $\operatorname{coker} d^P_{n+1}$ and $P_n$ are projective, then $B_n(P_\bullet)$ is also projective. Consequently $P'' = \cdots \mapsto P_{n+1} \mapsto B_n(P) \mapsto 0$ is K-projective.

Let the acyclic complex $N_\bullet$, then

$$0 \mapsto \operatorname{Hom}_A(\overline{P}_\bullet, N_\bullet) \mapsto \operatorname{Hom}_A(P_\bullet, N_\bullet) \mapsto \operatorname{Hom}_A(P''_\bullet, N_\bullet) \mapsto 0$$

is exact. And since $\operatorname{Hom}_A(P_\bullet, N_\bullet)$ and $\operatorname{Hom}_A(P''_\bullet, N_\bullet)$ are acyclic, then $\operatorname{Hom}_A(\overline{P}_\bullet, N_\bullet)$ is also acyclic. So $\overline{P}_\bullet$ is K-projective.

And $\forall i \leq n$, $H_i(P_\bullet) \cong H_i(\overline{P}_\bullet)$, and for $i > n$, $H_i(P_\bullet) \cong H_i(\overline{P}_\bullet) = 0$, so $H(P_\bullet) = H(\overline{P}_\bullet)$.

$5 \implies 6$ We know that $\tau_{\leq n} : P_\bullet \mapsto \overline{P}_\bullet$ is a quasi-isomorphism, so since $\overline{P}_\bullet$ is K-projective, there exists a quasi-isomorphism $\sigma : \overline{P}_\bullet \mapsto P_\bullet$ which splits the exact sequence $0 \mapsto P''_\bullet \mapsto P_\bullet \xmapsto{\tau_{\leq n}} \overline{P}_\bullet \mapsto 0$. Let $P'_\bullet = \operatorname{im}\sigma$ be K-projective. Consequently $P_\bullet \cong P'_\bullet \oplus P''_\bullet$ and $\forall i > n$, $P'_i = 0$. Since $P''_\bullet$ is acyclic and K-projective, it is then contractible.

$6 \implies 1$ From the direct sum $P_\bullet \cong P'_\bullet \oplus P''_\bullet$ we obtain the projection $\pi : P_\bullet \mapsto P'_\bullet$ which is a quasi-isomorphism. And since $\ker(\pi) = P''_\bullet$ is contractible, we have $P'_\bullet \simeq M_\bullet$. So $\operatorname{pd}_A(M_\bullet) \leq n$.

□

**Proposition 3.2.** *For any complex $N_\bullet$ the following conditions are equivalent:*

1. $\operatorname{id}_A(N_\bullet) \leq n$.
2. *For any complex $M_\bullet$, $\forall i > n + \sup M_\bullet$, $\operatorname{Ext}^i_A(M_\bullet, N_\bullet) = 0$*
3. *For any left ideal $J$ of $A$, $\operatorname{Ext}^{n+1}_A(A/J, N_\bullet) = 0$.*
4. *$\forall i < -n$, $H_i(N) = 0$ and for any K-injective resolution $I_\bullet$ of $M_\bullet$, $\ker d^I_{-n}$ is injective.*

5. *For any K-injective resolution $I_\bullet$ of $M_\bullet$, $\tau_{\geq -n}(I)$ is K-injective.*

6. *Any K-injective resolution $I_\bullet$ of $M_\bullet$ is isomorphic to a coproduct $I'_\bullet \oplus I''_\bullet$ such that $I'_\bullet = 0$, $\forall i < -n$ and $I''$ is contractible.*

**Proposition 3.3.** *For any complex $M_\bullet$ the following conditions are equivalent:*

1. *$fd_A(M_\bullet) \leq n$.*

2. *$\forall i > n$, $H_i(M_\bullet) = 0$ and for any K-flat resolution of $P_\bullet$ of $M_\bullet$, coker $(d^P_{n+1})$ is flat.*

3. *For any K-flat resolution $P_\bullet$ of $M_\bullet$, $\tau_{\leq n} : F_\bullet \mapsto \tau_{\leq n}(F_\bullet)$ is a quasi-isomorphism and $\tau_{\leq n}$ is K-flat.*

# Conclusion

In this document we observe that, to obtain a resolution or compute the homological dimension of a complex, one must impose certain boundedness conditions. Even in the unbounded case, the relevant constructions arise as infinite approximations of bounded ones. We still lack general properties that would allow us to express homological data for an arbitrary complex purely from its intrinsic structure. Achieving such results may require a substantial change of framework, or at least a fundamentally new perspective on the subject.

# Bibliography


[1] Luchezar L. Avramov and Hans-Bjørn Foxby. Homological dimensions of unbounded complexes. *Journal of Pure and Applied Algebra*, 71(2):129–155, 1991. Special Issue In Honor of H. Matsumura.

[2] Nicolas Bourbaki. *Éléments de mathématique. Algèbre commutative. Chapitres 1 à 4.* Berlin: Springer, reprint of the 1985 original edition, 2006.

[3] Saunders Mac Lane. *Homology.* Class. Math. Berlin: Springer-Verlag, reprint of the 3rd corr. print. 1975 edition, 1995.

[4] N. Spaltenstein. Resolutions of unbounded complexes. *Compositio Mathematica*, 65(2):121–154, 1988.

[5] Charles A Weibel. *An introduction to homological algebra.* Number 38. Cambridge university press, 1994.